

# Newcomb’s Paradox as a Postselected Closed Timelike Curve

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Newcomb’s paradox asks how an earlier forecast can remain reliably correlated with a later act. We model the forecast and the act as two qubits, linked by a postselection filter that keeps agreement histories more often than mismatch histories, by a ratio  $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$ . Read cyclically, this is the same final-state boundary condition used to model postselected closed timelike curves, though the calculations below use only that map and leave open whether a spacetime realization exists.  $\Gamma$  decides the ordinary decision problem: at an unbiased predictor prior, one-boxing pays more once the record’s distinguishability  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$  exceeds the reward ratio  $m/M$ , and an ordinary classical predictor with the same reliability gives the same verdict. Symmetrizing the forecast and the act gives two Newcomb problems side by side: the game is a Prisoner’s Dilemma at small  $\Gamma$ , and turns into a Stag Hunt once  $\Gamma$  exceeds  $(M + m)/M$ , with cooperation becoming the risk-dominant choice past  $\Gamma = (M + m)/(M - m)$ . The quantum content lies off the payoff basis: a  $\sigma_x$  coherence in the act qubit, bounded by a distinguishability–visibility tradeoff, marks the coherent part of the filter.

## I. INTRODUCTION

William Newcomb posed the problem; Nozick [1] brought it to philosophy, and Gardner’s *Scientific American* columns [2, 3] disseminated it widely. A transparent box contains a small reward  $m$ , and an opaque box contains either a large reward  $M$  or nothing. The agent chooses either to take only the opaque box, one-boxing, or to take both boxes, two-boxing. Before the choice is made, a predictor  $\Omega$  has already acted: it placed  $M$  in the opaque box if it predicted one-boxing, and left the box empty if it predicted two-boxing. The predictor is assumed highly accurate. Two-boxing appears to dominate once the boxes are prepared, because it adds the transparent reward whatever the opaque box contains; one-boxing appears better before conditioning on the prediction, because reliable one-boxers usually face a filled box. That tension is the familiar Newcomb dispute. The present paper isolates the physical issue underneath it: how can a forecast made earlier remain correlated with an act performed later?

The construction is a two-record P-CTC final-state map. Section II separates the selected-branch numerator from the normalized accepted-state update. The diagonal payoff and game-theoretic sections, Secs. III–V, use per-attempt utilities: success-probability-weighted conditional values of the normalized branch. The off-diagonal action gives the coherence test of Sec. VII. A further epistemic reading, developed in Sec. IX B, treats the same binary map as  $\Omega$ ’s consistency update over unresolved causal routes.

The familiar disagreement about what to do is a disagreement about how the record–act correlation is represented [4–8]. Wolpert and Benford [9] locate the issue in the probabilistic structure relating choice and prediction. Lewis [10] read a Prisoner’s Dilemma as a Newcomb problem, and Sobel [11] asked which Newcomb problems are Prisoner’s Dilemmas. We carry that classification one

type further. The postselected consistency filter of Sec. II supplies the correlating channel Lewis stipulated, and the strength of that channel fixes the game: a Prisoner’s Dilemma when the predictor is weak, a Stag Hunt when it is strong, with the crossover set by the reward ratio. Schmidt-Petri [12] reaches a one-boxing coordination verdict by extending Newcomb’s problem to repeated play. Here the single-shot filter separates the appearance of a coordination equilibrium,  $\Gamma = (M + m)/M$ , from the risk-dominance threshold,  $\Gamma = (M + m)/(M - m)$ .

We set the free-will question aside and model the act as a qubit  $X$ : its final computational-basis readout will be carried out as one-boxing or two-boxing. The “choice” is therefore a physical record produced at measurement. The predictor  $\Omega$  is granted every classical fact about the setup, the preparation of  $X$  included; the single thing denied it is the realized act itself. Classically,  $\Omega$  can read  $x$  and implement a policy  $y(x)$  without changing what  $x$  will be. With a decision qubit  $X$ , however, any mechanism that extracts the information needed to set  $Y$  is a measurement interaction in the  $X$  basis. It creates a record correlated with the eventual readout and therefore changes the state from which the later choice emerges. An early classical fact about the act is such a record; Assumption 1 (Sec. II) denies  $\Omega$  that record.

Once the predictor is forbidden from reading out  $X$  early, prediction consistency is enforced indirectly, by acting on the joint state of the relevant registers. The quantum system represents the predictor assumption as a joint constraint on  $X$  and  $Y$ . Newcomb’s predictor and the grandfather paradox then become two versions of the same physical question: how a theory represents a consistency constraint between a record and an event. In the grandfather paradox the inconsistent histories are those in which the event on the loop prevents its own cause; in Newcomb’s problem they are the mismatch histories, in which the predictor recorded one act and the agent later performs the other.

Closed-timelike-curve (CTC) models provide the language for that constraint. In a strong reading  $\Omega$  is a time traveler [13], a reference with its own closure: the tachyonic antitelephone is joint work of Newcomb himself and the Benford of [9]. Deutsch’s formulation [14] requires the state entering the loop to equal the state leaving it, a fixed-point condition on the chronology-violating system; for Newcomb-like situations the central object is the correlation between the loop system and external records, which the fixed-point map can wash out [16]. The postselected closed timelike curves (P-CTCs) of Lloyd et al. [15] take a different route: the would-be loop is implemented by quantum teleportation postselected on a fixed final-state outcome. The selected final-state projection supplies a pre-normalized branch; conditioning on it gives the nonlinear P-CTC state update.

The circuit of Fig. 1 is the corresponding two-record version of that final-boundary structure. The ancilla carries the consistency condition; read cyclically, it enforces a relation between the prior record and the later act. The calculations use the selected branch and its survival weights; the stronger claim that the same final-boundary map would be realized by an actual spacetime CTC is kept separate and discussed in Sec. IX. Postselection carries the computational power of  $\text{PostBQP} = \text{PP}$  [17], and whether a postselected consistency condition can be promoted from a circuit-level final-boundary map to a spacetime resource remains open [18].

A postselected parity check on the agent’s act  $X$  and  $\Omega$ ’s prior record  $Y$  retains agreement histories at rate  $p_{\text{match}}$  and disagreement histories at rate  $p_{\text{mismatch}}$ . Their odds ratio is the *consistency factor*,

$$\Gamma := \frac{p_{\text{match}}}{p_{\text{mismatch}}}, \quad (1)$$

which measures how strongly the selected ensemble favors agreement over mismatch in the raw record labels. At  $\Gamma = 1$  the postselection supplies no consistency bias. As  $\Gamma \rightarrow \infty$ , mismatch histories are eliminated and the map collapses to the agreement projector. Finite  $\Gamma > 1$  keeps a controlled mismatch rate, with relative weight  $\Gamma^{-1}$ . As a P-CTC, the filter is a two-sector consistency condition in the class treated by Ji et al. [19]. The payoff-ordering analysis depends on the filter only through  $\Gamma$  (Sec. II). When the mechanism is noisy, an unobserved spread in the filter angle  $\theta$  (Sec. II), equivalently dephasing of the ancilla about the rotation axis, averages to an effective consistency factor  $\Gamma_{\text{eff}}$ , obtained from the averaged agreement and disagreement survival weights.

The construction can be motivated by a reading of the short film *Stag Hunt* [25], much as Ji et al. [19] use *Interstellar* to illustrate retrocausal communication. In the reading used here, the two protagonists are paired records of one cyclic decision variable, each experienced from a different worldline. The binary labels may be read differently on the two worldlines: one line can code 0 as death and 1 as survival, while the other uses the inverted coding. Consistency is equality of physical content after

fixing the coding on each line. The filter of Sec. II assigns relative survival weights to the two possible closures, raw label agreement and raw label opposition. On this reading, the room’s isolation is the narrative counterpart of protecting the consistency factor from noise (Sec. VIII).

Two named  $2 \times 2$  games classify the payoff tables the filter induces. A normal-form game is a payoff table for two players, and a Nash equilibrium is a strategy profile at which no unilateral deviation raises the deviator’s payoff [29]. A strategy strictly dominates another when it gives a higher payoff against every strategy of the partner. In a Prisoner’s Dilemma, defection strictly dominates cooperation, so mutual defection is the unique Nash equilibrium, and it pays both players less than mutual cooperation. In a Stag Hunt [30], neither strategy dominates and both diagonal profiles are strict Nash equilibria; selecting between them is the equilibrium-selection problem of Harsanyi and Selten [31], taken up in Sec. IV B. Section IV folds the filter’s survival weights into the Newcomb payoff table, whose ordering changes type as  $\Gamma$  grows; Sec. V reads the result as two Newcomb problems set side by side.

The construction supports three results. First, at the unbiased setup prior the one-boxing verdict is classical and convention-independent: one-boxing is optimal once the distinguishability  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ , the record’s advantage over chance at inferring the act, exceeds the reward ratio  $m/M$ , matching the Wolpert–Benford resolution (Sec. III); at Gardner’s million-to-thousand stakes a predictor one part in a thousand better than a coin suffices. Second, under per-attempt scoring the consistency factor carries the induced symmetric game from Prisoner’s-Dilemma to Stag-Hunt ordering (Sec. IV). Third, prediction has a measurement-like tradeoff: the distinguishability and the act register’s transverse visibility  $\mathcal{V}$  obey  $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$ , with equality for the coherent implementation, and  $\Gamma \rightarrow \infty$  is represented as act-basis dephasing (Secs. II and VII); the coherent part is the off-diagonal  $\sigma_x$  diagnostic of Sec. VII A. This diagonal/off-diagonal split is the organizing principle of the paper.

Table I maps the regimes considered and the section treating each.

The paper follows the same split. Section II builds the parity filter and derives  $\Gamma$ . Section III fixes the scoring conventions and proves the convention-independent verdict at the unbiased prior. Section IV computes the induced payoff ordering and the Prisoner’s-Dilemma/Stag-Hunt thresholds; Sec. V gives the Lewis side-by-side reading. Section VI states the adaptive-predictor results, with derivations in Appendices H–J. Section VII derives the distinguishability–visibility bound and the  $\sigma_x$  signature, while Sec. VIII treats noise through  $\Gamma_{\text{eff}}$ . The remaining appendices contain the survival-weight reduction, conditional-on-success calculation, potential-landscape calculation, and process-matrix check.

| Regime           | prior $\beta$         | scoring  | key result                                            |
|------------------|-----------------------|----------|-------------------------------------------------------|
| Payoff ordering  | —                     | per att. | PD/SH at $\Gamma = \frac{M+m}{M}$ (Prop. E)           |
| Unbiased verdict | $\beta = \frac{1}{2}$ | both     | $\Gamma = \frac{M+m}{M-m}$ (Thm. 1)                   |
| Fixed- $\beta$   | exog. $\beta$         | per att. | boundary Eq. (21)                                     |
| Adaptive (fcst.) | $\beta_O(\alpha)$     | per att. | $\alpha^* = 1$ for $\Gamma > \frac{M+m}{M}$ (Sec. VI) |
| Adaptive (adv.)  | $\arg \min_\beta$     | per att. | $\Gamma = \frac{M+m}{m}$ ; Eq. (43)                   |
| Cond., fixed $Y$ | any                   | on succ. | entrywise two-box dom. (App. D)                       |
| Cond., policy    | $\beta$               | on succ. | prior-dep.; = per att. at $\beta = \frac{1}{2}$       |
| Noisy $\theta$   | —                     | per att. | $\Gamma \rightarrow \Gamma_{\text{eff}}$ , Eq. (49)   |

TABLE I. Map of regimes. Each row specifies a distinct decision setup through its setup-prior treatment and scoring convention. The setup-prior column is where the adaptive thresholds enter. Forward references are intentional; the table is a reading guide.

## II. A TWO-SECTOR POSTSELECTED CONSISTENCY CONDITION

The mechanism is a postselected consistency filter: register  $Y$  represents  $\Omega$ 's prior record and  $X$  the agent's later act, with the postselection weighting agreement histories  $Y = X$  against mismatch histories  $Y \neq X$ . In the ordinary Newcomb coding, mismatch is a forecast error. In the cyclic paired-worldline reading of Sec. IX, mismatch is a raw-label relation; the physical consistency condition is applied after fixing how the binary labels are read on the two worldlines. If  $\rho$  is the joint state and  $K$  is the Kraus operator for the successful final-state outcome, the P-CTC state update, on the nonzero-success domain, is

$$\rho \mapsto \frac{K\rho K^\dagger}{\text{Tr}(K\rho K^\dagger)}. \quad (2)$$

We will use three quantities attached to this selected branch:

$$\begin{aligned} \mathcal{N}_K(\rho) &:= K\rho K^\dagger, \\ p_K(\rho) &:= \text{Tr} \mathcal{N}_K(\rho), \\ \Phi_K(\rho) &:= \frac{\mathcal{N}_K(\rho)}{p_K(\rho)}. \end{aligned} \quad (3)$$

Here  $K$  is the successful-branch Kraus operator,  $\mathcal{N}_K$  is the linear trace-decreasing numerator,  $p_K(\rho)$  is the success probability, and  $\Phi_K$  is the normalized P-CTC state map of Eq. (2). The predictor's epistemic situation, stated informally in the introduction, is used throughout, so we record it once:

**Assumption 1** (the predictor's epistemic access).  $\Omega$  knows every classical fact about the setup, including the preparation of the act register. It does not possess the realized act, which is produced only at the final measurement.

Later sections invoke Assumption 1 wherever  $\Omega$  conditions on the agent's preparation (Secs. VI and VII).

### A. Circuit

An ancilla qubit  $a$  is prepared in  $|0\rangle$ . Two controlled-NOT (CNOT) gates write the parity  $x \oplus y$  into  $a$ ; a rotation  $R_y(\theta)$  follows, and the branch is kept only when  $a$  is measured as  $|0\rangle$ . The rotation is

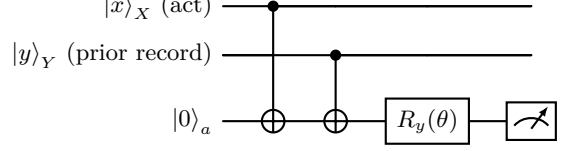


FIG. 1. The diagram represents the effective postselected map. The ancilla records the parity  $x \oplus y$ , is rotated by  $R_y(\theta)$ , and the branch is retained only when the ancilla outcome is  $|0\rangle$ . Agreement and disagreement histories carry survival weights  $\cos^2(\theta/2)$  and  $\sin^2(\theta/2)$ , respectively.

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \quad (4)$$

equivalently  $R_y(\theta) = e^{-i\theta\sigma_y/2}$ . After the two CNOTs, the ancilla carries the parity: it is  $|0\rangle$  on agreement and  $|1\rangle$  on disagreement. Postselecting immediately on  $a = |0\rangle$  eliminates mismatch histories entirely. The rotation  $R_y(\theta)$  makes the consistency check tunable: the selected outcome has overlaps  $\cos(\theta/2)$  and  $-\sin(\theta/2)$  with raw agreement and raw mismatch. Lloyd et al.'s postselected-teleportation P-CTC enforces consistency by conditioning on a selected Bell outcome [15]. Equation (A5) below gives the corresponding Bell contraction for this two-record filter. The selected final state defines the numerator  $\mathcal{N}_K$ , while normalization by  $p_K(\rho)$  gives Eq. (2). Incoherent noise during the would-be CTC channel can be represented as an unobserved spread in  $\theta$ ; for diagonal payoffs this replaces  $\Gamma$  by the finite factor  $\Gamma_{\text{eff}}$  of Sec. VIII.

### B. Acceptance probabilities

Let  $A$  (for “acceptance”) denote the event that the ancilla measurement returns 0, so that the branch is retained. The acceptance amplitude is  $\langle 0|R_y(\theta)|x \oplus y\rangle$ . Hence

$$\begin{aligned} p_{\text{match}}(\theta) &:= \mathbb{P}(A \mid x = y) = |\langle 0|R_y(\theta)|0\rangle|^2 \\ &= \cos^2(\theta/2), \end{aligned} \quad (5)$$

$$\begin{aligned} p_{\text{mismatch}}(\theta) &:= \mathbb{P}(A \mid x \neq y) = |\langle 0|R_y(\theta)|1\rangle|^2 \\ &= \sin^2(\theta/2). \end{aligned} \quad (6)$$

We refer to the pair  $(p_{\text{match}}, p_{\text{mismatch}})$  as the *sector survival weights* of the filter, the survival probabilities of the

agreement sector  $\mathcal{H}_=$  and disagreement sector  $\mathcal{H}_\neq$  under postselection. The match-to-mismatch ratio,

$$\Gamma := \frac{p_{\text{match}}}{p_{\text{mismatch}}} = \cot^2(\theta/2), \quad (7)$$

summarizes the strength of the filter.  $\Gamma$  is the consistency factor introduced in Eq. (1). The ordering results below depend on the mechanism only through  $\Gamma$ ; a different physical implementation with the same  $\Gamma$  induces the same normal form.

### C. Kraus operator for the postselected map

Conditioning on the ancilla outcome  $|0\rangle$  yields a trace-decreasing completely positive map on  $XY$  with Kraus operator  $K(\theta) = \langle 0|_a R_y(\theta) \text{CNOT}_{Y \rightarrow a} \text{CNOT}_{X \rightarrow a} |0\rangle_a$ . Evaluating on computational basis states gives

$$K(\theta) = \cos(\theta/2) (|00\rangle\langle 00| + |11\rangle\langle 11|) - \sin(\theta/2) (|01\rangle\langle 01| + |10\rangle\langle 10|). \quad (8)$$

For any input  $\rho_{XY}$ , the subnormalized postselected state is  $\tilde{\rho}_{XY} = K\rho_{XY}K^\dagger$  with success probability  $p = \text{Tr}(\tilde{\rho}_{XY})$ , and conditional state  $\rho'_{XY} = \tilde{\rho}_{XY}/p$ . This realizes the postselected update of Eq. (2). For inputs diagonal in the computational basis, the sign in Eq. (8) drops out: the successful branch only multiplies agreement probabilities by  $c^2$  and disagreement probabilities by  $s^2$ , with  $c = \cos(\theta/2)$  and  $s = \sin(\theta/2)$ . For coherent superpositions of the two parity sectors, the sign is observable. Cross-sector coherences acquire a factor  $-cs$ , so off-diagonal probes such as the  $\sigma_x$  measurement of Sec. VII A can distinguish the coherent filter from a classical rejection sampler with the same survival weights. Thus  $K(\theta)$  defines a trace-decreasing CP numerator  $\mathcal{K}_\theta(\rho) := K(\theta)\rho K^\dagger(\theta)$  on  $XY$ . This is a two-record instance of the CP-numerator-plus-renormalization structure studied in the retrocausal-channel literature [19, 26]. Interpreted cyclically,  $\mathcal{K}_\theta$  is a postselected mechanism with a final-state boundary condition on the ancilla, a reduced two-sector instance of the postselected-teleportation maps treated by Ji et al. [19]. Section VIII relates this CP numerator to their noisy-P-CTC formalism; the scalar  $\Gamma$  remains the agreement/mismatch survival ratio controlling the decision problem. The sector decomposition of the effect makes the correspondence concrete. Let

$$\Pi_+ := |00\rangle\langle 00| + |11\rangle\langle 11|, \quad \Pi_- := |01\rangle\langle 01| + |10\rangle\langle 10|. \quad (9)$$

The postselection effect is

$$E(\theta) := K^\dagger(\theta)K(\theta) = p_{\text{match}}\Pi_+ + p_{\text{mismatch}}\Pi_-. \quad (10)$$

Since  $E$  acts as  $p_{\text{match}}I$  on  $\mathcal{H}_=$  and  $p_{\text{mismatch}}I$  on  $\mathcal{H}_\neq$ , the consistency factor is  $\Gamma = \|E|_{\mathcal{H}_=}\|/\|E|_{\mathcal{H}_\neq}\| = p_{\text{match}}/p_{\text{mismatch}}$ , the likelihood ratio between the two sectors. Successful postselection multiplies the prior agreement odds by this ratio (Eq. (12)). The basis-diagonal

payoff analysis depends on  $E$  and on  $\Gamma$ ; the phase convention in  $K$  does not enter for computational-basis readouts. Off-diagonal observables such as the  $\sigma_x$  signature of Sec. VII A do depend on this phase.

The Bell-contraction realization of this operator is given in App. A. There a parity-dependent loop phase gives Bell overlaps  $\cos(\theta/2)$  on agreement and  $-\sin(\theta/2)$  on mismatch, reproducing Eq. (8); the endpoint  $K(0) = \Pi_+$  is the sharp P-CTC consistency check.

### D. Reliability in the accepted ensemble

Suppose  $X$  and  $Y$  are chosen independently with  $\Pr(X = 0) = \alpha$  and  $\Pr(Y = 0) = \beta$ . Before postselection,

$$\begin{aligned} \Pr(x = y) &= \alpha\beta + (1 - \alpha)(1 - \beta), \\ \Pr(x \neq y) &= \alpha(1 - \beta) + (1 - \alpha)\beta. \end{aligned}$$

Bayes' rule gives the agreement probability in the accepted ensemble:

$$\Pr(x = y | A) = \frac{p_{\text{match}} \Pr(x = y)}{p_{\text{match}} \Pr(x = y) + p_{\text{mismatch}} \Pr(x \neq y)}. \quad (11)$$

Equivalently, Bayes' rule takes the prior odds for agreement to

$$\frac{\Pr(x = y | A)}{\Pr(x \neq y | A)} = \Gamma \frac{\Pr(x = y)}{\Pr(x \neq y)}. \quad (12)$$

Equation (12) establishes the statistical meaning of  $\Gamma$ : it is the multiplicative evidence supplied by the consistency condition in favor of agreement. Small  $\theta$  makes  $p_{\text{mismatch}} \ll p_{\text{match}}$ , and the accepted ensemble is overwhelmingly consistent. At uniform input ( $\alpha = \beta = 1/2$ ) Eq. (11) reduces to  $\Pr(Y = X | A) = \Gamma/(\Gamma + 1)$ , so  $\Gamma = 1$  gives chance and  $\Gamma \rightarrow \infty$  gives certainty. Equivalently, the accepted ensemble assigns the record a *distinguishability*, the advantage over chance for inferring the act  $X$  from the record  $Y$ ,

$$\mathcal{D} := 2\Pr(Y = X | A) - 1 = \frac{\Gamma - 1}{\Gamma + 1} \quad (13)$$

at the unbiased preparation, a one-to-one function of  $\Gamma$  mapping  $[1, \infty)$  onto  $[0, 1)$ : at  $\mathcal{D} = 0$  the record is no better than a coin, at  $\mathcal{D} \rightarrow 1$  it fixes the act. Section III states the decision verdict in terms of  $\mathcal{D}$ , and Sec. VII returns to the same  $\mathcal{D}$  as the which-way distinguishability paired with transverse visibility.

### E. Basis-diagonal payoff comparisons depend only on $\Gamma$

*Proposition.* Let  $K$  define the selected branch  $\rho \mapsto K\rho K^\dagger$  on  $\mathcal{H}_X \otimes \mathcal{H}_Y = \mathbb{C}^2 \otimes \mathbb{C}^2$  and implement a parity check on  $(X, Y)$ : (i)  $K$  is diagonal in the computational



basis ( $K|xy\rangle = K_{xy}|xy\rangle$ ) and (ii) symmetric under joint bit-flip ( $(\sigma_x \otimes \sigma_x) \cdot K \cdot (\sigma_x \otimes \sigma_x) = K$ , with  $\sigma_x$  the Pauli bit-flip).

Then  $K = k_{\equiv}\Pi_{\equiv} + k_{\neq}\Pi_{\neq}$  with  $k_{\equiv} := K_{00} = K_{11}$  and  $k_{\neq} := K_{01} = K_{10}$ . The effect  $E = K^\dagger K$  equals  $p_{\text{match}}\Pi_{\equiv} + p_{\text{mismatch}}\Pi_{\neq}$  with  $p_{\text{match}} = |k_{\equiv}|^2$  and  $p_{\text{mismatch}} = |k_{\neq}|^2$ . For any product input  $\rho_X \otimes \rho_Y$  diagonal in the  $\{C, D\}$  basis and any payoff observable diagonal in the same basis, the binary policy comparison between any two preparations  $\alpha, \alpha'$  depends on  $K$  only through  $\Gamma := p_{\text{match}}/p_{\text{mismatch}}$ .

*Proof.* See Appendix B.  $\square$

The proposition underlies the convention-independent verdict (Sec. III). The basis-diagonal hypothesis is necessary: the  $\sigma_x$  observable of Sec. VII A is not basis-diagonal, so the proposition does not constrain it, and its  $\Gamma$ -dependence separates the coherent construction from a classical rejection sampler.

The decision analysis may then be read where agreement passes losslessly:  $K = \Pi_{\equiv} + \Gamma^{-1/2}\Pi_{\neq}$  with  $p_{\text{match}} = 1$ ,  $p_{\text{mismatch}} = 1/\Gamma$ , only disagreement histories discarded. The symmetric realization  $R_y(\theta)$  carries the same  $\Gamma$  and supplies the coherence signature of Sec. VII A and the noise model of Sec. VIII.

The parity decomposition separates the analysis. Secs. III–V and Sec. VI read the diagonal, where every payoff comparison is classical and a rejection sampler at matching survival weights reproduces it. Sec. VII reads the off-diagonal coherence, and Sec. VIII tracks the noise that degrades each sector.

### III. SCORING CONVENTIONS AND THE INVARIANT VERDICT

The consistency condition acts on histories before any payoff is read, and its strength is set by  $\Gamma$  alone (Sec. II). The scoring conventions differ in whether the success probability of the selected final-state branch is retained.

*Conditional* scoring evaluates the payoff after the P-CTC normalization,

$$\bar{U}_{\text{cond}}(\rho) = \text{Tr}[U \Phi_K(\rho)] = \frac{\text{Tr}[U \mathcal{N}_K(\rho)]}{p_K(\rho)}. \quad (14)$$

*Per-attempt* scoring is the success-weighted value of the same conditional branch,

$$\bar{U}_{\text{att}}(\rho) = p_K(\rho) \text{Tr}[U \Phi_K(\rho)] = \text{Tr}[U \mathcal{N}_K(\rho)]. \quad (15)$$

The act readout is part of  $U$ : conditional scoring reads it on the normalized branch  $\Phi_K(\rho)$ , while per-attempt scoring also counts how often that branch is reached. The two sectors carry Newcomb's two kinds of outcome. Match ( $x = y$ ) is where  $\Omega$ 's record agrees with the act. Mismatch ( $x \neq y$ ) is the case Newcomb's perfect-predictor stipulation eliminates:  $\Omega$ 's record disagrees with the act,

so the two-boxer finds the opaque box filled and the one-boxer finds it empty. The postselection plays the role of Newcomb's predictor: the mismatch scenarios it would eliminate survive only at relative rate  $\Gamma^{-1}$ . The accepted set then contains every match scenario together with a  $\Gamma^{-1}$  fraction of mismatch scenarios, the latter vanishing in the perfect-predictor limit  $\Gamma \rightarrow \infty$ . If a joint profile  $(x, y)$  carries acceptance weight  $p_{xy}$  (so  $p_{xy} = p_{\text{match}}$  when  $x = y$  and  $p_{xy} = p_{\text{mismatch}}$  when  $x \neq y$ ), the per-attempt payoff is

$$u_i^{\text{eff}}(x, y) = p_{xy} u_i(x, y), \quad (16)$$

the unnormalized (consistency-weighted) expectation. It keeps the suppression of inconsistent histories attached to the payoff, which is what makes the temptation profile rare per attempt.

The diagonal thresholds below depend only on these sector weights. A rejection sampler with the same weights reproduces them; the complementary-basis test of Sec. VII is what distinguishes coherent enforcement. At the unbiased setup prior, Theorem 1 makes  $p_K(\rho)$  preparation-independent, so the conditional and per-attempt comparisons have the same sign. Away from that locus the denominator in Eq. (2) matters, and the conventions separate. The game-theoretic structure of Secs. IV and V is a theorem about Eq. (15); Appendix D gives the conditional calculation.

#### A. Base payoffs

Let  $m$  denote the transparent-box reward and  $M$  the potential opaque-box reward, with  $M \gg m$  in the standard Newcomb setup [3]. Write  $x \in \{C, D\}$  for the agent's action (one-box  $C$  vs two-box  $D$ ), and  $y \in \{C, D\}$  for the prior-record variable (the predictor's setup). The choosing role receives the standard Newcomb payoffs

$$\begin{aligned} u_1(C, C) &= M, & u_1(D, C) &= M + m, \\ u_1(C, D) &= 0, & u_1(D, D) &= m. \end{aligned} \quad (17)$$

We label  $Y$  so that  $y = C$  means *predict*  $X = C$  (opaque box filled) and  $y = D$  means *predict*  $X = D$  (opaque box empty).

#### B. Ensemble payoff at a fixed prior

The per-attempt utility is an ensemble statistic: the average payoff over the statistical ensemble of histories generated by the postselected joint distribution. Writing the ensemble as  $N$  samples, the accumulated payoff is

$$W_N = \sum_{k=1}^N A_k u(x, y_k), \quad (18)$$

where  $A_k \in \{0, 1\}$  is the postselection outcome on sample  $k$ . The ratio  $W_N/N$  is the ensemble average, the same

single-shot expectation  $\text{Tr}[U K \rho K^\dagger]$  defined above, now read as an ensemble mean. The ensemble is a statistical device for evaluating that expectation; no sequence of physical trials is consumed.

Let

$$\beta := \Pr(Y = C)$$

be the preselection probability that the predictor's record is  $C$  before the consistency filter acts. The accuracy that defines Newcomb's predictor is carried by  $\Gamma$ , and  $\beta$  sets only the record's base rate:  $\beta = 1/2$  encodes an unbiased record, and a strongly accurate predictor is a large  $\Gamma$  at any  $\beta$ . This subsection treats  $x$  as a fixed classical policy throughout the ensemble. Section VI relaxes that assumption: when the preparation is a decision qubit and  $\beta$  responds to it, the answer depends on  $\Omega$ 's objective, with accuracy-seeking, calibrated, and adversarial predictors giving different thresholds.

For a one-boxer ( $x = C$  on every run), only the  $y = C$  branch contributes:

$$\bar{u}(C) = \beta p_{\text{match}} M. \quad (19)$$

For a two-boxer ( $x = D$ ),

$$\bar{u}(D) = \beta p_{\text{mismatch}}(M + m) + (1 - \beta) p_{\text{match}} m. \quad (20)$$

One-boxing has higher ensemble average iff  $\bar{u}(C) > \bar{u}(D)$ .

**Proposition** (Prior-dependent payoff boundary). *For preselection setup prior  $\beta \in (0, 1]$  and  $M > m > 0$ , one-boxing has the larger ensemble-average payoff per attempt iff*

$$\Gamma > \frac{\beta(M + m)}{\beta M - (1 - \beta)m}, \quad (21)$$

*provided  $\beta M > (1 - \beta)m$ ; otherwise one-boxing is never payoff-optimal, even in the perfect-consistency limit.*

The condition  $\beta M > (1 - \beta)m$  is the viability condition for one-boxing in this ensemble comparison: if it fails, even the perfect-consistency limit  $\Gamma \rightarrow \infty$  does not make one-boxing have the larger ensemble-average payoff. At the unbiased setup prior  $\beta = 1/2$  the boundary (21) reduces to  $\Gamma > (M + m)/(M - m)$ . The theorem below shows that this verdict is convention-independent, and Sec. IV B identifies the same threshold with risk dominance of the one-boxing equilibrium.

### C. The invariant verdict at the unbiased prior

At  $\beta = 1/2$  the agent's preparation has no effect on the acceptance rate, and the two scoring conventions return the same verdict.

**Theorem 1** (acceptance invariance and a convention-independent verdict). *Fix  $M > m > 0$  and  $\Gamma \neq 1$ . Prepare the act register with one-boxing Born weight  $\alpha = \Pr(X = C)$  and the prior record with  $\Pr(Y = C) = \beta$ , and let the parity filter act with sector survival weights  $(p_{\text{match}}, p_{\text{mismatch}})$ ,  $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$ .*

- (i) (Acceptance invariance.) *The per-attempt acceptance probability is the total probability of success over the agreement and disagreement sectors. Since*

$$\Pr(x = y \mid \alpha, \beta) = \alpha\beta + (1 - \alpha)(1 - \beta), \quad \Pr(x \neq y \mid \alpha, \beta) = \alpha(1 - \beta)$$

*one has*

$$\Pr(A \mid \alpha, \beta) = p_{\text{match}} [\alpha\beta + (1 - \alpha)(1 - \beta)] + p_{\text{mismatch}} [\alpha(1 - \beta) + (1 - \alpha)\beta].$$

*Hence*

$$\frac{\partial \Pr(A \mid \alpha, \beta)}{\partial \alpha} = (p_{\text{match}} - p_{\text{mismatch}})(2\beta - 1).$$

*At  $\beta = 1/2$ ,  $\Pr(A \mid \alpha, 1/2) = (p_{\text{match}} + p_{\text{mismatch}})/2$  for every  $\alpha$ , so the agent's preparation exerts no influence on the acceptance rate.*

- (ii) (Convention independence.) *At  $\beta = 1/2$  the per-attempt comparison  $\bar{u}(C)$  versus  $\bar{u}(D)$  and the conditional-on-success comparison  $U_A(C)$  versus  $U_A(D)$  (the expected utility given acceptance, App. D) have the same sign, both favoring one-boxing iff*

$$\Gamma > \frac{M + m}{M - m}.$$

- (iii) (Decision-theoretic match.) *With accepted-ensemble reliability  $q = \Pr(Y = X \mid A) = \Gamma/(\Gamma + 1)$ , the boundary in (ii) is  $q > (M + m)/2M$ , matching the Wolpert-Benford resolution at symmetric forecast accuracy.*

*Proof.* Part (i) is the law of total probability applied to the two parity sectors, followed by the displayed differentiation. Part (ii) compares the per-attempt and conditional payoffs at  $\beta = 1/2$ . Part (iii) substitutes  $q = \Gamma/(\Gamma + 1)$  into the threshold. See App. C.  $\square$

*Remark* (why  $\beta = 1/2$  is the unbiased setup). At  $\beta = 1/2$  the acceptance rate  $(p_{\text{match}} + p_{\text{mismatch}})/2$  is independent of the agent's preparation, so the two scoring conventions agree (part i). The one-boxing advantage above  $(M + m)/(M - m)$  then comes entirely from the record-act correlation aligning the agent's payoff with the surviving sector. For  $\beta \neq 1/2$  the preparation moves the acceptance rate, the conventions separate, and a Simpson reversal, a sign flip of the comparison upon conditioning, appears (Sec. IX D). We take  $\beta = 1/2$  as the symmetric reference case: there Nozick's agent cannot change the contents of an already-filled box, and an asymmetric  $\beta$  introduces a base-rate bias the original setup does not stipulate.

The boundary of Theorem 1 has a compact statement in the accepted-ensemble distinguishability of Eq. (13), the record's advantage over chance for inferring the later act from the earlier record. At  $\beta = \frac{1}{2}$ ,

$$\text{one-box} \iff \mathcal{D} > \frac{m}{M}, \quad (22)$$

so the one-boxing payoff is larger when the accepted-ensemble distinguishability exceeds the reward ratio. At the classic million-to-thousand stakes ( $m/M = 10^{-3}$ ) this threshold is  $\mathcal{D} = 0.1\%$ , so a predictor only 1.048596% distinguishable from a coin flip would clear it tenfold. The payoff-ordering threshold of Sec. IV sits just below it at  $\mathcal{D} = m/(2M + m)$ , and the adversarial adaptive threshold of Sec. VI at  $\mathcal{D} = M/(M + 2m)$ ; Fig. 4 places the three on the  $\mathcal{D}$  axis.

The verdict is classical. By the proposition of Sec. II E, every quantity entering the comparison depends on the mechanism only through the sector survival weights, and a rejection sampler that keeps agreement profiles with probability  $p_{\text{match}}$  and disagreement profiles with probability  $p_{\text{mismatch}}$  reproduces all of them. The quantum content lives off the payoff basis, a payoff-independent probe of whether the consistency condition was realized coherently; Sec. VII locates it.

#### D. Relation to the Wolpert–Benford resolution

Wolpert and Benford (WB) resolve Newcomb’s paradox by showing that the two conflicting recommendations correspond to two probabilistic structures relating the player’s choice  $x$  and the predictor’s forecast  $y$  [9]. Their Fearful structure decomposes  $P(x, y) = P(y | x) P(x)$ , treating forecast accuracy  $P(y | x)$  as given; their Realist structure decomposes  $P(x, y) = P(x | y) P(y)$  with  $P(y)$  fixed by the predictor and the agent’s strategy a  $y$ -independent conditional  $P(x | y) = h(x)$ , so the choice does not depend on the forecast. The two cannot be merged: specifying both conditional families simultaneously overconstrains the joint (§3.5 of [9]). WB rest their resolution on this incompatibility and set the accuracy parameter aside; we reinstate it here to surface the threshold it controls within their Fearful structure.

The accepted-ensemble reliability  $q := \Pr(Y=X | A) = \Gamma/(\Gamma + 1)$  (Sec. II D) maps onto Fearful’s symmetric accuracy parameter. At  $\Gamma = 1$ ,  $q = 1/2$ : the postselection induces no record–act dependence, and the accepted joint is the product distribution, the Realist assignment. As  $\Gamma \rightarrow \infty$ ,  $q \rightarrow 1$ : the accepted joint approaches Fearful’s perfect-predictor limit  $P(y | x) = \delta_{y,x}$ .

Reinstating a symmetric accuracy  $q$  in the Fearful structure, the two values are

$$\begin{aligned} V_{\text{one}}(q) &= Mq, \\ V_{\text{two}}(q) &= (M + m)(1 - q) + mq = M + m - Mq, \end{aligned} \quad (23)$$

so one-boxing wins iff  $q > (M + m)/(2M)$ . Within the Fearful structure WB note that the optimal choice is determined once the accuracy is fixed (their §3.1), but treat its value as immaterial to dissolving the paradox; the boundary made explicit here is that accuracy threshold. Converting the threshold of Theorem 1 through  $q =$

$\Gamma/(\Gamma + 1)$ :

$$q > \frac{(M + m)/(M - m)}{(M + m)/(M - m) + 1} = \frac{M + m}{2M},$$

the same condition; Sec. IV B identifies it with risk dominance of the one-boxing equilibrium.

The lower payoff-ordering threshold  $(M + m)/M$ , where the Stag-Hunt equilibrium first appears (Sec. IV), is additional game structure that their pure decision-theory treatment does not resolve; the consistency factor supplies this finer threshold as well.

The construction does not merge the Fearful and Realist structures: it parameterizes the Fearful family through the postselection collider and meets the Realist joint only at  $\Gamma = 1$ . No single value of  $\Gamma$  instantiates both decompositions simultaneously, so the no-merge theorem is not violated. That the decision content is entirely classical is consistent with their conclusion that quantum mechanics is not essential to the paradox. The two decompositions are the two time directions of the conditional. Realist conditions the act on a fixed prior forecast, the forward direction: the act cannot depend on it, and the joint is the product at  $\Gamma = 1$ . Fearful conditions the forecast on the act, the backward direction, with the record tracking the later choice once  $\Gamma > 1$ . Their time-reversal-invariance result (§3.4 of [9]), that the paradox is unchanged if prediction occurs after the choice, supports the reading-relative treatment of the collider/common-cause contrast in Sec. IX D (Fig. 2).

#### IV. THE INDUCED GAME: FROM PRISONER’S DILEMMA TO STAG HUNT

Under per-attempt scoring, the sector weights of the selected P-CTC branch enter through the success-weighted value in Eq. (15). The resulting normal form changes ordinal type as  $\Gamma$  grows. The game is a phase diagram for the diagonal survival weights: the distinguishability  $\mathcal{D}$  changes the ordering of the effective payoff observable, with the thresholds of Sec. III reappearing as equilibrium boundaries. The agreement filter is symmetric between the two registers, so the induced normal form is symmetric, and we use the standard  $2 \times 2$  notation

$$\begin{array}{c|cc} & C & D \\ \hline C & (R, R) & (S, T) \\ D & (T, S) & (P, P) \end{array} \quad (24)$$

to classify the effective payoff ordering. The symmetric table is a bookkeeping device for the two-register payoff ordering: the register  $Y$  is still  $\Omega$ ’s prior record, and equilibrium-selection language becomes literal only once  $Y$  is promoted to a second decision register (Sec. V).

The symmetrized payoff table has Prisoner’s-Dilemma ordering when

$$T > R > P > S,$$

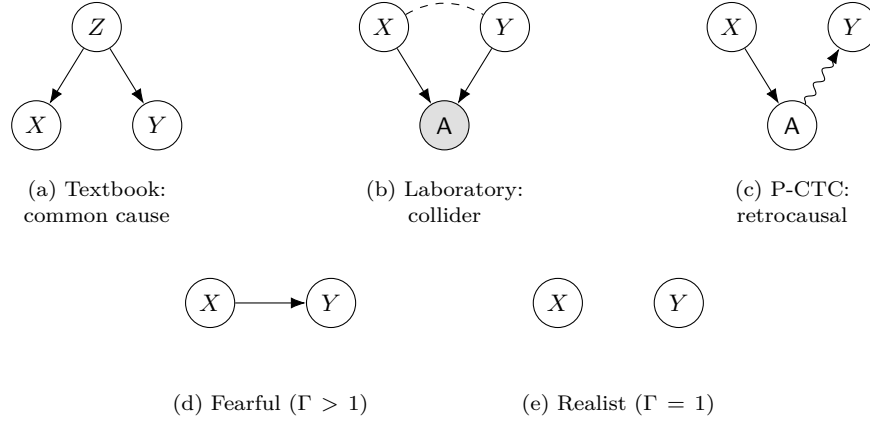


FIG. 2. Causal readings of the choice–forecast relation. Top row: a common-cause reading (a), a collider reading in which conditioning on the success event  $A$  induces the  $X$ – $Y$  association (b), and the cyclic P-CTC reading in which the final boundary is read as influence on the prior record (c). These instantiate the Reichenbach alternatives: common cause or direct cause, with the collider as the non-causal association [27, 28]. Bottom row: the Wolpert–Benford Fearful factorization, forecast conditioned on act, and the Realist factorization, independence at  $\Gamma = 1$ . The verdict is reported at the symmetric prior, where the scoring conventions agree.

so  $D$  strictly dominates  $C$  and  $(D, D)$  is the unique diagonal fixed point (a Nash equilibrium of the symmetric game, Sec. V). It has the Stag-Hunt structure [30] when

$$R > T, \quad R > P, \quad P > S,$$

so both  $(C, C)$  and  $(D, D)$  are diagonal fixed points.

If one further distinguishes strict ordinal payoff chains, the coordination region splits into two subregimes:  $R > T > P > S$  is the strict ordinal Stag-Hunt chain, and  $R > P > T > S$  is a strong-assurance, or diagonal-dominant, coordination regime, with the same two diagonal fixed points and both matched outcomes dominating both mismatched outcomes.

At  $\Gamma = 1$ , the postselection gives no consistency bias and the two registers are weighted symmetrically. As  $\Gamma$  increases, the survival factors raise  $R$  relative to  $T$ , and the ordering passes from  $T > R$  (Prisoner’s Dilemma) to  $R > T$  (Stag Hunt) at a threshold value of  $\Gamma$ . These inequalities classify the matrix only; equilibrium selection requires the Newcomb payoffs and the setup prior (Sec. IV B).

For each computational-basis profile  $(x, y)$ , the postse-

lection effect supplies the survival weight

$$p_{xy} = \langle xy | E(\theta) | xy \rangle = \begin{cases} p_{\text{match}}, & x = y, \\ p_{\text{mismatch}}, & x \neq y. \end{cases}$$

The effective normal-form entries are therefore the original payoff entries multiplied by these sector weights, as in Eq. (16).

#### A. Success-weighted Newcomb payoffs

To put the two roles on symmetric footing, define the second payoff by swapping arguments,

$$u_2(x, y) = u_1(y, x). \quad (25)$$

The swapped payoff  $u_2$  makes the match and mismatch sectors appear as a symmetric normal form;  $Y$ ’s role in that matrix is set by the analysis of Sec. VI. Under the per-attempt convention (16), each payoff entry of Eq. (17) is multiplied by the appropriate success probability. Using Eq. (5), the effective payoff matrix is

|     |                                                             |                                                             |      |
|-----|-------------------------------------------------------------|-------------------------------------------------------------|------|
|     | $C$                                                         | $D$                                                         |      |
| $C$ | $(p_{\text{match}}M, p_{\text{match}}M)$                    | $(p_{\text{mismatch}} \cdot 0, p_{\text{mismatch}}(M + m))$ | (26) |
| $D$ | $(p_{\text{mismatch}}(M + m), p_{\text{mismatch}} \cdot 0)$ | $(p_{\text{match}}m, p_{\text{match}}m)$                    |      |

Comparing with Eq. (24) identifies

$$\begin{aligned} R &= p_{\text{match}}M, & P &= p_{\text{match}}m, \\ T &= p_{\text{mismatch}}(M + m), & S &= 0. \end{aligned} \quad (27)$$

The temptation payoff  $M + m$  lives off-diagonal in the Newcomb table: it is the reward for choosing  $D$  when the prior-record variable is  $C$ . In the present mechanism that payoff is carried by a mismatch profile, hence inherits the



acceptance probability  $p_{\text{mismatch}}$ .

With these entries the symmetrized normal form changes ordinal type as  $\Gamma$  grows: it is Prisoner's-Dilemma-ordered for  $\Gamma < (M+m)/M$  and Stag-Hunt/assurance-ordered for  $\Gamma > (M+m)/M$ , and within the coordination region  $(C, C)$  is risk-dominant for  $\Gamma > (M+m)/(M-m)$ . Above  $\Gamma = (M+m)/m$  a strong-assurance regime sets in,  $R > P > T > S$ , where per attempt mutual defection outpays the unilateral temptation because the temptation branch is a suppressed mismatch. Appendix E gives the full ordinal classification with the degenerate endpoints and the proof.

These two boundaries are distinct, the second larger. In the interval between them the symmetrized table is Stag-Hunt-ordered, two diagonal fixed points exist, and the recommendation at  $\beta = 1/2$  remains two-boxing because  $(D, D)$  is risk-dominant there (Theorem 1). The recommendation changes at the risk-dominance boundary, so the equilibrium and its selection arrive at separate thresholds. This interval is region II of Fig. 3.

The nontrivial condition  $R > T$  reads

$$p_{\text{match}}M > p_{\text{mismatch}}(M+m), \quad (28)$$

equivalently

$$\Gamma > \frac{M+m}{M}. \quad (29)$$

When this condition holds, a second diagonal fixed point appears:  $(C, C)$ , the mutual one-boxing outcome. The temptation branch remains available, but it must pass through the mismatch sector, and the postselection can make it rare per attempt. The threshold sits a distance  $m/M$  above  $\Gamma = 1$ , so the signal strength required to flip the ordering is the reward ratio  $m/M$  itself.

### B. Basin threshold and risk dominance

Once the Newcomb payoffs have been inserted, the condition  $R > T$  marks the appearance of the one-boxing agreement equilibrium  $(C, C)$ . The setup prior  $\beta$  of Sec. III then determines on which side of the Stag-Hunt basin boundary the preparation lies. Choosing  $C$  has per-attempt expected payoff  $U(C|\beta) = \beta R + (1-\beta)S$ , and choosing  $D$  has  $U(D|\beta) = \beta T + (1-\beta)P$ ; the indifference prior is therefore  $\beta_{\text{ind}} = (P-S)/[(P-S) + (R-T)]$ . For  $\beta > \beta_{\text{ind}}$ , the one-boxing branch has the larger per-attempt value.

For the success-weighted Newcomb matrix,  $S = 0$ . Substituting Eq. (27) gives

$$\beta_{\text{ind}} = \frac{P}{P+R-T} = \frac{\Gamma m}{(M+m)(\Gamma-1)}. \quad (30)$$

Equation (30) is the setup-prior boundary between the one-boxing and two-boxing basins. Its endpoints give

the two thresholds:  $\beta_{\text{ind}} = 1$  gives  $\Gamma = (M+m)/M$ , where the one-boxing diagonal first appears as an equilibrium;  $\beta_{\text{ind}} = 1/2$  gives  $\Gamma = (M+m)/(M-m)$ , the risk-dominance boundary.

The latter condition is the Harsanyi–Selten risk-dominance criterion [31]: in a symmetric  $2 \times 2$  coordination game the risk-dominant equilibrium is the unique best reply to a partner mixing uniformly over the two equilibrium strategies, equivalently the equilibrium whose basin exceeds half the prior interval,  $\beta_{\text{ind}} < 1/2$ . The agreement equilibrium  $(C, C)$  is risk-dominant when

$$R - T > P - S.$$

Using Eq. (27), this becomes

$$p_{\text{match}}(M-m) > p_{\text{mismatch}}(M+m), \quad (31)$$

equivalently

$$\Gamma > \frac{M+m}{M-m}.$$

**Corollary** (Unbiased setup prior). *At  $\beta = 1/2$ , the per-attempt expected payoffs are proportional to the symmetrized normal-form payoffs. Hence risk dominance and the single-agent comparison of Theorem 1 coincide:  $(C, C)$  is risk-dominant iff one-boxing has the larger ensemble-average payoff per attempt.*

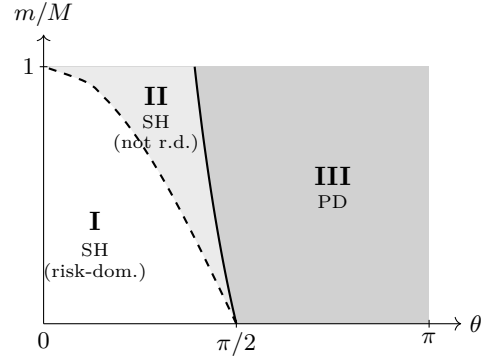


FIG. 3. Phase diagram in the  $(\theta, m/M)$  plane, with  $\Gamma = \cot^2(\theta/2)$ . **I** (white): Stag-Hunt payoff ordering with one-boxing  $(C, C)$  risk-dominant. **II** (light gray): Stag-Hunt payoff ordering, but two-boxing  $(D, D)$  is risk-dominant. **III** (dark gray): Prisoner's-Dilemma ordering, where  $D$  strictly dominates. The solid curve is the payoff-ordering boundary (29); the dashed curve is the risk-dominance boundary (31). Both curves meet at  $\theta = \pi/2$  in the limit  $m/M \rightarrow 0$ . The plotted range is  $0 < m/M < 1$ , corresponding to  $M > m > 0$ .

Figure 3 shows the two boundaries in the  $(\theta, m/M)$  plane. The solid curve is the payoff-ordering boundary  $R = T$ . The dashed curve is the risk-dominance boundary  $R - T = P - S$ ,

$$\theta_{\text{rd}} = 2 \arctan \left[ \left( \frac{1 - m/M}{1 + m/M} \right)^{1/2} \right],$$

and it lies inside the Stag-Hunt region. As  $\theta$  increases from 0, the postselection filter becomes less selective: region I has Stag-Hunt payoff ordering and risk-dominant agreement, region II has Stag-Hunt payoff ordering with  $(D, D)$  risk-dominant, and region III has Prisoner's-Dilemma ordering, with  $D$  strictly dominant. For  $M \gg m$  the two curves lie close to each other near  $\theta = \pi/2$ , their separation set by  $m/M$ ; for Gardner's values the two thresholds are  $\Gamma_{\text{SH}} = 1.001$  and  $\Gamma_{\text{RD}} \approx 1.002$ . That reweighting any Prisoner's Dilemma by a strong enough diagonal-success correlation yields coordination is generic; what is specific here is that a single physical parameter  $\Gamma = \cot^2(\theta/2)$  sets both thresholds and ties them to the reward ratio  $m/M$ .

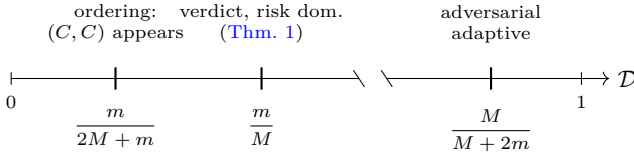


FIG. 4. Threshold ladder on the distinguishability axis  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$ . The one-boxing equilibrium appears at  $m/(2M + m)$ , the verdict and risk-dominance switch at  $m/M$ , and a nonzero adversarial one-boxing weight only above  $M/(M + 2m)$ . For Gardner's values these are approximately  $5.0 \times 10^{-4}$ ,  $10^{-3}$ , and 0.998.

Figure 4 places the three thresholds on the distinguishability axis: the ordering and verdict thresholds are both of order  $m/M$ , while the adversarial adaptive threshold sits within  $2m/M$  of perfect distinguishability. Section VI quantifies the separation.

*Potential and basins.* The symmetrized table (26) is an exact potential game whose two coordination attractors are separated by an interior repeller at  $\beta_{\text{ind}}$ , the deeper well switching to one-boxing at the risk-dominance boundary  $\Gamma = (M + m)/(M - m)$ ; App. F gives the potential, barrier, and replicator dynamics. In Landau terms the double well makes the Prisoner's-Dilemma-to-Stag-Hunt boundary  $\Gamma = (M + m)/M$  the metastability onset, where the second minimum appears, and the risk-dominance boundary  $\Gamma = (M + m)/(M - m)$  the first-order transition, where the deeper well switches, the one-boxing weight serving as order parameter. The equilibrium-selection content is the static counterpart of an attractor structure.

## V. LEWIS'S SIDE-BY-SIDE READING

This section tests whether Lewis's side-by-side identification survives beyond the Prisoner's-Dilemma regime. Lewis [10] read a Prisoner's Dilemma as two Newcomb problems set side by side: each player treats the partner's correlated action as a forecast of their own. The parity filter supplies that correlation explicitly. Eq. (26) already contains the needed payoff table. The point of the side-by-side reading is that the single-agent and two-player

questions are distinct: holding the partner's act fixed tests the raw Newcomb dominance relation, while putting two such problems together exposes the equilibrium structure of the composite.

First, each side remains a Newcomb problem. Promote  $Y$  to a second decision register and fix player  $j$ 's act as the forecast of player  $i$ 's act. The raw Newcomb dominance relation is unchanged,

$$u_i(D, x_j) - u_i(C, x_j) = m, \quad x_j \in \{C, D\}, \quad (32)$$

while the accepted ensemble supplies the reliability

$$q = \Pr(x_j = x_i | A) = \frac{\Gamma}{\Gamma + 1} \quad (33)$$

at the symmetric profile. Hence the parametric comparison on either side is

$$V_C(q) = Mq, \quad V_D(q) = M + m - Mq, \quad (34)$$

so the Newcomb verdict turns to one-boxing at

$$\Gamma = \frac{M + m}{M - m}. \quad (35)$$

Second, the payoff ordering changes before the Newcomb verdict/risk-dominance boundary. The side-by-side normal form becomes Stag-Hunt ordered at

$$\Gamma = \frac{M + m}{M}, \quad (36)$$

where the second diagonal fixed point  $(C, C)$  appears. The interval

$$\frac{M + m}{M} < \Gamma < \frac{M + m}{M - m} \quad (37)$$

is therefore the gap between the appearance of  $(C, C)$  and the risk-dominance switch: the one-boxing equilibrium exists, but  $(D, D)$  is still risk-dominant. At Eq. (35), the side-by-side Newcomb comparison and the Harsanyi-Selten risk-dominance comparison cross at the same value.

Lewis's Prisoner's-Dilemma case is thus the weak-predictor end of a one-parameter family. Increasing the same consistency factor that raises the predictor's reliability carries the side-by-side composite into Stag Hunt, without changing the raw Newcomb dominance relation on either side. The lower threshold has no single-agent counterpart: it is the point at which a second diagonal equilibrium appears in the composite. Section IX returns to a cyclic reading, in which the second decision register is interpreted as the same agent on a parallel worldline. Appendix G gives the formal proposition.

## VI. ADAPTIVE PREPARATION: THE PREDICTOR'S OBJECTIVE

The fixed-prior comparison of Sec. III treats the preparation of  $Y$  as external. By Assumption 1,  $\Omega$  knows

the agent's preparation;  $\beta$  may therefore respond to it, and the adaptive verdict depends on what  $\Omega$  optimizes. Newcomb's original predictor is closer in spirit to an accuracy-seeking forecaster than to an adversary. We therefore separate three objectives, holding the postselection mechanism, and hence  $\Gamma$ , fixed. The derivations are collected in Appendices H–J.

Let the decision qubit be prepared as

$$|\psi(\alpha)\rangle_X = \sqrt{\alpha} |C\rangle + \sqrt{1-\alpha} |D\rangle,$$

so that the final computational-basis readout gives  $C$  with Born weight  $\alpha$  and  $D$  with Born weight  $1-\alpha$ . The realized value of  $X$  is still not read out before the postselection. Knowing  $\alpha$ ,  $\Omega$  chooses the prior-record preparation

$$\rho_Y(\beta) = \beta |C\rangle\langle C| + (1-\beta) |D\rangle\langle D|.$$

The agent's decision is the preparation  $\alpha$ , a probability of one-boxing; the act  $X$  is its downstream Born-rule readout. A relative phase in the preparation of  $X$  leaves the calculation unchanged, because the postselection effect, the prior-record state, and the payoff observable are all diagonal in the  $\{|C\rangle, |D\rangle\}$  basis.

For a general symmetric  $2 \times 2$  normal form with entries  $(R, S, T, P)$ , the per-attempt ensemble payoff is

$$W(\alpha, \beta) = \alpha\beta R + \alpha(1-\beta)S + (1-\alpha)\beta T + (1-\alpha)(1-\beta)P. \quad (38)$$

Equivalently,

$$W(\alpha, \beta) = P + \alpha(S - P) + \beta(T - P) + \alpha\beta\Delta, \quad (39)$$

$$\Delta = R - S - T + P.$$

Thus

$$\frac{\partial W}{\partial \alpha} = S - P + \beta\Delta, \quad \frac{\partial W}{\partial \beta} = T - P + \alpha\Delta.$$

For the Newcomb entries (27),  $\Delta = (M+m)(p_{\text{match}} - p_{\text{mismatch}})$ , so  $\Delta > 0$  for  $\Gamma > 1$  and  $\Delta = 0$  at  $\Gamma = 1$ .

The three adaptive objectives give sharply different thresholds.

**Forecasting predictor.** If  $\Omega$  maximizes forecast accuracy, the maximum-a-posteriori (MAP) forecast is the threshold rule

$$\beta_{\text{MAP}}(\alpha) = \mathbf{1}[\alpha > \tfrac{1}{2}], \quad (40)$$

with the tie at  $\alpha = \frac{1}{2}$  assigned to  $\beta = 0$ . Optimizing against this response gives

$$\alpha_{\text{MAP}}^* = 1 \quad \text{for all } \Gamma > \frac{M+m}{M}. \quad (41)$$

Thus, for an accuracy-seeking predictor, pure one-boxing is the payoff-maximizing preparation throughout the Stag-Hunt region. In the Prisoner's-Dilemma region the value is only a supremum as  $\alpha \rightarrow \frac{1}{2}^+$ : the agent is forecast as

a one-boxer while keeping as much two-boxing weight as possible (Appendix H).

**Calibrated predictor.** If  $\Omega$  sets  $\beta = \alpha$ , then  $W(\alpha, \alpha)$  is convex in  $\alpha$  for  $\Gamma > 1$ , so the optimum is at a corner. Since  $R > P$ ,

$$\alpha_{\text{cal}}^* = 1 \quad \text{for all } \Gamma \geq 1. \quad (42)$$

The same verdict holds for a self-calibrated predictor satisfying  $\beta = \Pr(X = C | \mathbf{A})$  (Appendix I). Functional decision theory recommends one-boxing [7]; the calibrated column returns that recommendation as an output of the postselection mechanism, one-boxing being selected independently of the mechanism's threshold structure whenever the predictor is calibrated.

**Adversarial predictor.** If  $\Omega$  selects  $\beta$  to minimize the agent's payoff, the robust adaptive value is the lower envelope

$$V_{\text{adv}}(\alpha) := \inf_{\beta \in [0,1]} W(\alpha, \beta).$$

The robust optimum is

$$\alpha_{\text{adv}}^* = \begin{cases} 0, & \Gamma \leq \frac{M+m}{m}, \\ \frac{\Gamma m - (M+m)}{(M+m)(\Gamma-1)}, & \Gamma > \frac{M+m}{m}. \end{cases} \quad (43)$$

The corresponding dual prior in the mixed regime is

$$\beta^* = \frac{\Gamma m}{(M+m)(\Gamma-1)}, \quad \beta^* - \alpha^* = \frac{1}{\Gamma-1}. \quad (44)$$

As  $\Gamma \rightarrow \infty$ , both  $\alpha^*$  and  $\beta^*$  tend to  $m/(M+m)$ . Even perfect consistency therefore leaves the lower-envelope preparation overwhelmingly two-boxing in the standard Newcomb regime  $M \gg m$ . For Gardner's values  $M = 10^6$ ,  $m = 10^3$ , the adversarial threshold is 1001, and the ideal-limit one-boxing weight is  $9.99 \times 10^{-4}$  (Appendix J).

No additional P-CTC assumption is introduced here: the postselected map contributes only the sector survival weights, and adaptive preparation changes the input ensemble fed to the same trace-decreasing map. With noise, the formulas hold with  $\Gamma$  replaced by the effective consistency factor  $\Gamma_{\text{eff}}$  (Sec. VIII).

## VII. THE PREDICTOR'S PREDICAMENT: A DISTINGUISHABILITY-VISIBILITY COMPLEMENTARITY

Once the act is the computational-basis readout of a decision qubit, the early access denied to the predictor by Assumption 1 is bounded by a complementarity relation: the predictor's accuracy (the distinguishability  $\mathcal{D}$ ) and the act register's visibility are complementary quantities, constrained by an inequality the coherent implementation meets with equality. This locates the construction's

| Objective    | response $\beta_O(\alpha)$         | opt. $\alpha$                 | onset                                            |
|--------------|------------------------------------|-------------------------------|--------------------------------------------------|
| MAP forecast | $\mathbf{1}[\alpha > \frac{1}{2}]$ | 1                             | $C$ for $\Gamma > \frac{M+m}{M}$                 |
| Calibrated   | $\alpha$                           | 1                             | $C$ for all $\Gamma \geq 1$                      |
| Adversarial  | $\arg \min_{\beta} W$              | $0 \rightarrow \frac{m}{M+m}$ | nonzero $C$ -weight for $\Gamma > \frac{M+m}{m}$ |

TABLE II. Optimal decision-qubit preparation under three predictor objectives, same postselection filter throughout. The MAP predictor is the Newcomb-faithful objective; the adversarial column is a robustness bound.

quantum content in the predictor’s inability to acquire the act as an early classical fact, while the decision itself stays classical (Sec. III).

Keep the decision qubit  $|\psi(\alpha)\rangle_X = \sqrt{\alpha}|C\rangle + \sqrt{1-\alpha}|D\rangle$  of Sec. VI, with the act the outcome  $x \in \{C, D\}$  of the later readout. For this section widen the predictor to its broadest form: any process that, before the final act readout, couples  $X$  to a record system  $Y$ , whose computational-basis readout  $y$  is the predictor’s forecast. Let

$$q := \Pr(y = x) \quad (45)$$

be its reliability in the sense of Sec. IID, now for an arbitrary record  $y$ ,  $\mathcal{D} := 2q - 1 \in [0, 1]$  the corresponding *distinguishability*, extending Eq. (13) to that record ( $\mathcal{D}$ , distinct from the action label  $D$ ), and

$$\mathcal{V} := \sqrt{\langle \sigma_x \rangle_X^2 + \langle \sigma_y \rangle_X^2}$$

the phase-optimized transverse visibility of the act register, its surviving coherence in the plane complementary to the payoff basis. Equivalently, if  $\rho_{CD} := \langle C|\rho_X|D\rangle$ , then  $\mathcal{V} = 2|\rho_{CD}|$ : the “fringe” is the phase response of the two-state act register, obtained by inserting a relative phase between  $C$  and  $D$  and reading in the complementary basis. In the real-phase  $R_y(\theta)$  implementation used below, this visibility lies in the  $\sigma_x$  quadrature, so  $\mathcal{V} = |\langle \sigma_x \rangle_X|$ . For the parity-filter family, the predictor regime is  $q \geq 1/2$ , equivalently  $\Gamma \geq 1$  ( $\theta \in [0, \pi/2]$ ); for  $\Gamma < 1$  the record is anti-correlated with the act, and  $\mathcal{D}$  is taken after optimal relabeling. Fig. 3 shows  $\Gamma < 1$  only to complete the ordering classification.

**Proposition** (Distinguishability–visibility complementarity). Let  $X$  be the act qubit and  $Y$  a record system with a designated computational-basis readout  $y$ , with  $q = \Pr(y = x)$ , distinguishability  $\mathcal{D} := 2q - 1 \in [0, 1]$ , and phase-optimized transverse visibility  $\mathcal{V}$ . Then

$$\mathcal{D}^2 + \mathcal{V}^2 \leq 1, \quad (46)$$

with equality in the pure optimal-record case used below. *Proof.* This is the standard two-alternative distinguishability–visibility argument. The  $X$ -basis alternatives play the role of paths, and  $Y$  is the record whose readout supplies which-alternative information. Englert’s distinguishability is defined after optimizing the record measurement; the fixed readout bias  $\mathcal{D} = 2q - 1$

used here cannot exceed that optimized distinguishability. The Englert/Jaeger–Shimony–Vaidman bound therefore gives  $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$ . The equality conditions are checked in App. K.  $\square$

The parity filter realizes the bound. At the symmetric preparation  $\alpha = \beta = 1/2$ , the coherent filter  $K = c\Pi_{=} + e^{i\varphi}s\Pi_{\neq}$  has accepted-ensemble distinguishability  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$  and visibility  $\mathcal{V} \leq 2\sqrt{\Gamma}/(\Gamma + 1)$ . The real-phase circuit of Fig. 1 places this visibility entirely in the  $\sigma_x$  quadrature and meets the bound with equality,

$$\mathcal{D} = \frac{\Gamma - 1}{\Gamma + 1} = \cos \theta, \quad \mathcal{V} = \sin \theta, \quad \mathcal{D}^2 + \mathcal{V}^2 = 1. \quad (47)$$

The accepted state  $|\psi(\theta)\rangle$  is pure and the computational-basis readout of  $Y$  is the optimal which-way measurement in the two-alternative problem, the two conditions for equality in (47) (App. K). A dephased implementation with the same sector survival weights gives  $\mathcal{V} = 0$ ; the sector survival weights fix  $\mathcal{D}$  but leave the transverse coherence implementation-dependent.

**Corollary** (No early classical fact). For a genuine superposition  $0 < \alpha < 1$ , any record that predicts the act with certainty ( $q = 1$ , equivalently  $\Gamma \rightarrow \infty$ ) forces  $\mathcal{V} = 0$ : the act register is dephased to  $\rho_X = \alpha|C\rangle\langle C| + (1-\alpha)|D\rangle\langle D|$ , with no coherence in the complementary basis. Perfect prediction is therefore represented by dephasing in the act basis: the record fixes the later act readout only by removing the complementary coherence. Any mechanism that kept both perfect prediction and transverse coherence would fall outside the diagonal common-cause process of App. M; the present filter never produces such cross-register terms. The record–act consistency the predictor requires can be imposed only as a joint constraint on  $(X, Y)$ , which is what the postselected map  $K\rho K^\dagger$  does. *Proof.* With  $\Pr(y = x) = 1$  the joint readout is supported on the diagonal, so tracing out  $Y$  leaves  $\rho_X$  diagonal and  $\mathcal{V} = 0$ ; the endpoint  $\Gamma \rightarrow \infty$  of (47) realizes it. See App. K.  $\square$

The consistency factor  $\Gamma$  fixes  $\mathcal{D}$  and leaves  $\mathcal{V}$  free: the same sector survival weights admit a coherent implementation,  $\mathcal{V} = \sqrt{1 - \mathcal{D}^2}$ , and a dephased one,  $\mathcal{V} = 0$ , and the visibility distinguishes the two. The choice recommendation is carried by  $\mathcal{D}$  alone. The visibility is the complementary diagnostic of implementation, registering coherent enforcement of the consistency condition rather



than a classical reliability oracle. It is maximal at  $\Gamma = 1$ , where  $\mathcal{D} = 0$ , and zero at the corner  $(\mathcal{D}, \mathcal{V}) = (1, 0)$ , the perfect-prediction limit  $\Gamma \rightarrow \infty$ , the Omega point at which  $\Omega$  never errs. Figure 5 summarizes the geometry.

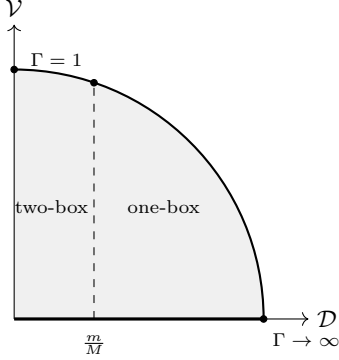


FIG. 5. Decision and coherence content in the  $(\mathcal{D}, \mathcal{V})$  plane at the symmetric preparation. Physical records lie in  $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$ ; the coherent implementation traces the unit arc, while a dephased one lies on  $\mathcal{V} = 0$ . The Newcomb verdict reads only the horizontal coordinate, one-boxing for  $\mathcal{D} > m/M$ ; the visibility is maximal at  $\Gamma = 1$  and vanishes in the perfect-prediction limit.

#### A. The $\sigma_x$ signature at the symmetric preparation

The basis-diagonal decision content of Secs. III–VI is classical: a rejection sampler at matching sector survival weights reproduces all expected utilities. The construction does, however, leave a basis-off-diagonal diagnostic whose visibility, by the bound (46), is complementary to the decision variable itself. For the real-phase  $R_y(\theta)$  realization, the visibility lies in the  $\sigma_x$  quadrature, and the expectation on  $X$  at  $\alpha = \beta = 1/2$  is  $\langle \sigma_x \rangle_X = -2\sqrt{\Gamma}/(\Gamma + 1)$ , while the dephased classical model gives zero (Table III).

Anything that distinguishes the construction from this classical model must live off-diagonal in  $\{C, D\}$ ; the uniqueness proposition of Sec. IIE covers basis-diagonal observables and does not extend to  $\sigma_x$ , which is why it can separate the coherent and dephased implementations. At the symmetric preparation  $\alpha = \beta = 1/2$  both registers are in  $\sigma_x$  eigenstates  $|+\rangle = (|C\rangle + |D\rangle)/\sqrt{2}$ . Sec. II noted that  $R_y(\theta)$  on the ancilla supplies the non-triviality at the postselection; the symmetric preparation places the registers themselves on  $\sigma_x$  eigenstates as well, and the  $\sigma_x$  observable on  $X$  is sensitive to the off-diagonal part of the selected numerator.

Carrying the  $|+\rangle_X |+\rangle_Y$  input through the two CNOTs, the ancilla rotation, and postselection on  $|0\rangle_a$  gives a pure joint state  $|\psi(\theta)\rangle_{XY}$  (App. K), with acceptance probability  $1/2$  independent of  $\theta$  and accepted match/mismatch weights  $\cos^2(\theta/2), \sin^2(\theta/2)$  that recover  $\Gamma = \cot^2(\theta/2)$ . Its reduced state on  $X$  is  $\rho_X(\theta) = \frac{1}{2}(I - \sin\theta \sigma_x)$ , so  $\langle \sigma_x \rangle_X = -2\sqrt{\Gamma}/(\Gamma + 1)$ . The sign is negative because

agreement-favoring postselection retains the branch whose Kraus amplitude on the mismatch sector carries the relative minus sign of Eq. (8), imprinting an anti-aligned  $\sigma_x$  coherence. The classical rejection-sampling model gives  $\rho_X^{\text{cl}} = I/2$  and  $\langle \sigma_x \rangle_X^{\text{cl}} = 0$  at every  $\Gamma$ , so this observable distinguishes the coherent construction from any classical rejection sampler with the same computational-basis sector survival weights.

The sector survival weights fix every basis-diagonal payoff comparison; the off-diagonal sector carries the remaining quantum content, whose local act-register signature depends on the implementation phase and is maximal for the real-phase  $R_y(\theta)$  circuit used here (App. K).

The  $\sigma_x$  signature is strongest at  $\Gamma = 1$ , where  $\langle \sigma_x \rangle_X = -1$ : at the indifference point neither sector is reweighted, the filter maps  $|+\rangle |+\rangle$  to  $|-\rangle |-\rangle$ , and the coherence is carried through at full strength with its sign reversed, a maximum at the point where the distinguishability vanishes ( $\mathcal{D} = 0$ ). At the endpoints the joint state stays Bell-coherent ( $|\Phi^+\rangle$  as  $\Gamma \rightarrow \infty$ ,  $|\Psi^+\rangle$  as  $\Gamma \rightarrow 0$ ) while the reduced state on  $X$  becomes  $I/2$  and the local signature vanishes. By exchange symmetry  $\langle \sigma_x \rangle_Y = \langle \sigma_x \rangle_X$ .

The symmetric preparation has decision-theoretic significance: at  $\beta = 1/2$  the per-attempt and conditional conventions agree (Theorem 1); risk dominance there coincides with the single-agent expected-payoff comparison (Sec. IV B). The  $\sigma_x$  signature is off-strategic: pure two-boxing ( $\alpha = 0$ ) and pure one-boxing ( $\alpha = 1$ ) leave the registers in computational-basis states with  $\langle \sigma_x \rangle = 0$ , so it lies in the complement of the payoff basis and leaves the strategic verdict untouched (Eq. (46)). The coherence maximum at  $\Gamma = 1$  sits in the PD-ordered region where the verdict is two-boxing, and its vanishing at the ideal-P-CTC limit is where Nozick’s perfect-predictor stipulation is realized.

As a diagnostic, compare the coherent  $|+\rangle_X |+\rangle_Y$  input with a dephased mixture matched on the computational-basis statistics. The basis-diagonal payoffs agree; only the coherent numerator carries the predicted  $\sigma_x$  expectation. The two descriptions separate only through this off-diagonal term, a coherence question distinct from the decision verdict. Under the noisy-channel reading of Sec. VIII,  $\Gamma_{\text{eff}}$  need not capture all cross-sector coherences, so the diagnostic assumes sharp control of  $\theta$ . Appendix K records a corresponding two-qubit postselection benchmark and the scaling of the selected-event count.

## VIII. NOISE AND THE EFFECTIVE CONSISTENCY FACTOR

A fixed, precisely known angle  $\theta$  gives an exact, noiseless map at every consistency strength  $\Gamma = \cot^2(\theta/2)$ , unitary loop interaction and all (App. A). In the time-travel reading, noise in the P-CTC channel is instead imperfect control of  $\theta$  itself: its value is treated as an unobserved ensemble variable rather than one known setting. A distribution of  $\theta$  averages to the same two-sector form with

| quantity at $\alpha = \beta = \frac{1}{2}$ | coherent $R_y(\theta)$         | dephased              |
|--------------------------------------------|--------------------------------|-----------------------|
| heralding $\Pr(A)$                         | 1/2                            | 1/2                   |
| accepted agreement $q$                     | $\Gamma/(\Gamma + 1)$          | $\Gamma/(\Gamma + 1)$ |
| visibility $\mathcal{V}$                   | $\sin \theta$                  | 0                     |
| $\langle \sigma_x \rangle_X$               | $-2\sqrt{\Gamma}/(\Gamma + 1)$ | 0                     |

TABLE III. Coherent parity filter against a dephased model matched on the computational-basis sector survival weights, at the symmetric preparation. The basis-diagonal entries ( $\Pr(A)$ ,  $q$ ) agree by construction; the transverse coherence ( $\mathcal{V}$ ,  $\langle \sigma_x \rangle_X$ ) separates the two. The contrast is largest at  $\Gamma = 1$ , where  $\langle \sigma_x \rangle_X = -1$  against 0.

survival ratio  $\Gamma_{\text{eff}}$ . For basis-diagonal payoff comparisons with unobserved  $\theta$ , only the averaged agreement and disagreement survival weights enter. What a resolved value of  $\theta$  would add is taken up at the end of this section.

Averaging the accepted-branch numerator  $K(\theta)(\cdot)K^\dagger(\theta)$  over the angle noise yields an effect of the same two-sector form as Eq. (10), because  $E(\theta) = K^\dagger K$  is diagonal in the sector basis for each  $\theta$  and diagonal matrices average to diagonal matrices:

$$E_{\text{eff}} = \mathbb{E}[\cos^2(\theta/2)] \Pi_{\pm} + \mathbb{E}[\sin^2(\theta/2)] \Pi_{\mp}. \quad (48)$$

The corresponding effective consistency factor is

$$\Gamma_{\text{eff}} = \frac{\mathbb{E}[\cos^2(\theta/2)]}{\mathbb{E}[\sin^2(\theta/2)]}. \quad (49)$$

Averaging over the angle noise places the single- $\theta$  numerator of Sec. II in the noisy-P-CTC framework of Ji et al. [19] with an explicit channel. Their noisy P-CTC is a noiseless P-CTC followed by a channel  $\mathcal{N}_\ell$ , entering their map through its Choi state  $\mathcal{N}_\ell[\Phi]$ . For the basis-diagonal decision content the averaged filter is this construction with  $\mathcal{N}_\ell$  the bit-flip channel

$$\mathcal{N}_\ell(\sigma) = (1 - \ell)\sigma + \ell\sigma_x\sigma\sigma_x, \quad \ell = \mathbb{E}[\sin^2(\theta/2)], \quad (50)$$

whose flip rate is the mean mismatch weight  $\ell$ : agreement survives at  $1 - \ell$ , disagreement at  $\ell$ , and the survival ratio is  $\Gamma_{\text{eff}} = (1 - \ell)/\ell$ , Eq. (49). Physically this channel is dephasing of the ancilla about the rotation axis: averaging  $R_y(\theta)$  shrinks the ancilla's  $\sigma_z$  coordinate, the parity  $x \oplus y$ , by  $1 - 2\ell$ , partially randomizing the match/mismatch sector that the postselection reads. Every basis-diagonal Newcomb threshold of Secs. IV–VI then follows from the sharp case by replacing  $\Gamma$  with  $\Gamma_{\text{eff}}$ : the ideal agreement-projector limit is  $\Gamma_{\text{eff}} \rightarrow \infty$  ( $\ell \rightarrow 0$ ), and  $\Gamma_{\text{eff}} = 1$  ( $\ell = \frac{1}{2}$ ) carries no match-mismatch consistency signal.

The boundaries are mismatch tolerances in  $\ell$ : Stag-Hunt ordering holds for  $\ell < M/(2M + m)$ , risk dominance of the agreement equilibrium for  $\ell < (M - m)/(2M)$ , and the adaptive-preparation verdict for  $\ell < m/(M + 2m)$ . For  $M \gg m$  the first two tolerances are close to  $1/2$  while the adaptive one is of order  $m/M$ , so the adaptive

condition is what tests whether the consistency signal stays sharp enough to move the agent's preparation.

Since  $\theta$  is a rotation angle, its noise wraps around the circle, where the von Mises distribution of concentration  $\kappa$  plays the role a Gaussian does on the line. It gives the mismatch weight in closed form:

$$\ell = \mathbb{E}[\sin^2(\theta/2)] = \frac{1}{2}(1 - A(\kappa)), \quad A(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \quad (51)$$

with  $A$  the mean resultant length and  $I_\nu$  the modified Bessel functions. The adaptive verdict survives while  $\ell < m/(M + 2m)$ , that is  $A(\kappa) > 1 - 2m/(M + 2m)$ . At Gardner's values  $M = 10^6$ ,  $m = 10^3$  this is  $\kappa \gtrsim 251$ . In the concentrated limit von Mises approaches a wrapped Gaussian of variance  $\sigma^2 = 1/\kappa$ , where  $\ell \simeq \sigma^2/4$  and the threshold reads  $\sigma < 2\sqrt{m/(M + 2m)} \sim 2\sqrt{m/M}$ , about  $0.063$  rad or  $3.6^\circ$ . The concentration is the sharpness of the consistency condition; in a heralded realization it is the control precision of the optical rotation.

The opposite extreme,  $\theta$  uniform on  $[0, \pi]$ , is the maximal-ignorance limit. At fixed  $\theta$  the ordering splits close to evenly: Stag-Hunt with probability  $\Pr(\text{SH}) = \theta^*/\pi$ ,  $\theta^* = 2 \arctan \sqrt{M/(M + m)}$ , leaving a small Prisoner's-Dilemma excess  $\Pr(\text{PD}) - \Pr(\text{SH}) = m/(\pi M) + O(m^2/M^2)$  set by the reward ratio. The averaged game is nonetheless Prisoner's-Dilemma-ordered,  $\Gamma_{\text{eff}} = 1$ : the payoff ordering is nonlinear in the sector weights, so the ordering of the average departs from the fixed- $\theta$  majority (App. L).

Dephasing the registers removes the  $\sigma_x$  coherence of Sec. VII A while leaving the diagonal survival weights of the classical rejection sampler (Sec. III D). Angle noise changes the match and mismatch survival weights and therefore replaces  $\Gamma$  by  $\Gamma_{\text{eff}}$  within the same diagonal reduction. The von Mises concentration  $\kappa$  of Eq. (51) fixes the effective distinguishability in closed form,  $\mathcal{D}_{\text{eff}} = A(\kappa)$ .

## IX. DISCUSSION

The preceding sections used one final-state map in two ways. The normalized map in Eq. (2) is the accepted-state P-CTC update. Per-attempt payoffs are its success-probability-weighted values, equal to the numerator expression in Eq. (15). The causal analysis and quantum-game comparison are collected below; the optional epistemic and relational readings are in Secs. IX B and IX C, and the process-matrix check is App. M.

### A. P-CTC numerator and spacetime realization

The P-CTC identification is a claim about the final-state map. The selected final state defines the Kraus operator  $K$ , the numerator  $\mathcal{N}_K$ , the success probability  $p_K(\rho)$ , and the normalized branch  $\Phi_K$ . The payoff-ordering results use  $p_K(\rho) \text{Tr}[U\Phi_K(\rho)]$ , equivalently the

sector weights of  $\mathcal{N}_K$ ; the coherence result uses the same Kraus map and the normalized accepted state. A space-time CTC realization would add an ontological claim beyond the calculations, which require only the final-boundary map [37].

This distinction is also the one used in black-hole final-state proposals [38]: the linear numerator is fixed by a final-state boundary condition, and the physical state is obtained only after renormalization. Here  $\Gamma$  sets the strength of that boundary condition, maximally mixed at  $\Gamma = 1$  and rank-reduced as  $\Gamma \rightarrow \infty$ . Ji et al. [19] limit retrocausal communication through such maps under arbitrary final-state boundary conditions, these models included, whose unitarity Lloyd and Preskill examine [39] and whose interactions with infalling matter can break it [40].

Promoting the final-boundary map to a genuine space-time resource is the open physical question. A geometric substrate is a Lorentzian metric with closed time-like curves (the Gödel universe [41], traversable wormholes [42], Gott’s cosmic strings [43]); whether such a metric can form is the content of the chronology protection conjecture [44], which holds that semiclassical backreaction forbids closed timelike curves. Novikov self-consistency imposes a consistent physical identification around such curves [45]. In the paired-worldline reading, that identification is Eq. (52): raw agreement and raw opposition are two label conventions for matching the records. The calculations do not depend on choosing a spacetime embedding.

Two features of postselected CTCs bear on the decision reading. First, Eq. (15) is linear because it restores the success probability that Eq. (2) divided out; per-attempt utility uses the pre-conditioning ensemble. Second, postselection confers the computational power of  $\text{PostBQP} = \text{PP}$  [17]; the open question is whether the postselected consistency condition can be realized as a spacetime resource or as a final-state simulation [18]. The  $\sigma_x$  diagnostic is conditioned on the success flag. This separates it from the stronger state-discrimination effects associated with Deutsch-type CTCs [46]. For postselected CTCs the corresponding discrimination power is weaker: linearly independent states can be distinguished, but linearly dependent sets cannot [47].

Process-matrix membership turns on normalization. A postselected numerator  $\mathcal{E}$  gives the conditional map  $\rho \mapsto \mathcal{E}(\rho)/\text{Tr } \mathcal{E}(\rho)$ ; this map is linear on a preparation family only when the success probability  $\text{Tr } \mathcal{E}(\rho)$  is constant on that family. Araújo, Guérin and Baumeler identify process matrices with a linear special case of P-CTC computation, bounded by  $\text{BQP}_{\text{CTC}} \subseteq \text{PP}$  [48]. In the present model, Theorem 1(i) gives constancy only on the diagonal preparation family at  $\beta = 1/2$ , where  $\partial \text{Pr}(\mathbf{A})/\partial \alpha = 0$ . Accordingly, App. M uses the accepted diagonal slice as the process-matrix object and leaves the arbitrary-instrument postselected update at the CP-map level. At the symmetric point this reduced object is positive, valid, no-signaling, and causally separable, so the payoff-level

records admit a common-cause embedding. The  $\sigma_x$  test distinguishes coherent implementation from causal non-separability of that reduced process.

The same map has two readings. Drawn as a forward laboratory circuit,  $Y$  becomes correlated with  $X$  through joint postselection. Drawn as a P-CTC, the numerator is a final-state boundary condition, so the same map is read cyclically as a constraint between an earlier record and a later act. Wolpert and Benford prove the paradox time-reversal invariant: it is unchanged whether  $\Omega$  forecasts before the choice or observes it after (§3.4 of [9]). Temporal priority carries none of the problem’s force, which lies in the structure relating record and act; the consistency filter supplies it. The record’s counterfactual dependence on the act, without backward causation, is how Craig vindicates one-boxing in the foreknowledge case [51]. A spacetime CTC or retrocausal interpretation can recover the temporal gloss; the payoff and coherence results require only the postselected consistency map.

Conditional scoring treats the agent as already in the normalized branch of Eq. (2); per-attempt scoring values that branch before conditioning on its occurrence. Theorem 1 shows that the Newcomb verdict at the unbiased prior is insensitive to this choice, because the P-CTC denominator is then preparation-independent on the relevant diagonal family.

## B. Epistemic consistency reading

The same formalism also admits an epistemic reading, where postselection represents  $\Omega$ ’s internal consistency update over possible causal routes. The predictor’s model may be arbitrarily large: it may include psychological variables, physical cues, memories, simulations, self-referential hypotheses, and further hidden structure. Newcomb’s problem exposes only a binary interface to that model, the forecast record  $Y \in \{C, D\}$  and the later act  $X \in \{C, D\}$ . Integrating out the internal variables leaves an effective two-sector description, agreement  $X = Y$  and disagreement  $X \neq Y$ , with likelihood ratio  $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$ .

In this sense the consistency filter is an effective theory of the predictor at the exposed binary interface. If  $\lambda$  labels internal causal routes and  $\pi(\lambda) = (x, y)$  is the coarse-graining to the Newcomb variables, the diagonal update only records how  $\Omega$  weights the two coarse-grained sectors. All basis-diagonal payoff comparisons are therefore unchanged: a coherent model, a classical rejection sampler, and a physical heralded filter agree whenever they have the same  $\Gamma$ .

The off-diagonal sector carries the extra content. In an epistemic reading, coherence should not be identified simply with ignorance about a hidden classical act. More cautiously, it can be read as the phase-sensitive part of a model in which the act has not yet been externalized as a public fact. A diagonal mixture over  $C$  and  $D$  describes uncertainty about an already classical alternative; a co-

herent superposition describes an unresolved act whose alternatives remain phase-related before externalization. Thus even a predictor with perfect knowledge of all settled classical facts need not have a determinate act-value to read off before the act is made public.

Measurement is then the interaction by which  $\Omega$ 's internal model becomes an external record. The complementarity relation  $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$  says that public predictive reliability and coherent openness cannot both be maximal: the more  $\Omega$  turns an unresolved act into a reliable forecast, the less complementary visibility can remain. The diagonal decision threshold can be read without committing to a spacetime loop. The P-CTC contribution is the final-state consistency map; the epistemic version below reads the same consistency update as a coherent response to an act not yet fixed as a public record.

### C. Relational reading

This subsection is only an interpretive gloss; none of the payoff, game-theoretic, process-matrix, or coherence results depends on relational quantum mechanics. In RQM, quantum states and values are not observer-independent properties of a system, but descriptions relative to another physical system after an interaction [20]. On that reading, the predictor's record  $Y$  is a fact relative to  $\Omega$ , while the later act  $X$  is a fact relative to the agent. The parity filter does not make those facts absolute. It only enforces, in the heralded ensemble, a controlled relation between the two records: match histories survive with weight  $\cos^2(\theta/2)$  and mismatch histories with weight  $\sin^2(\theta/2)$ , so  $\Gamma = \cos^2(\theta/2)/\sin^2(\theta/2)$  is the odds by which agreement is selected.

This vocabulary is useful because it separates two notions that the main text keeps apart. The diagonal statistic  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$  is the stable, basis-readable part of the record-act relation: it is the part a classical rejection sampler can reproduce and the only part entering the payoff comparisons of Sec. III. The visibility  $\mathcal{V}$  is the complementary coherent part. In the symmetric coherent preparation, the accepted act register has

$$\langle \sigma_x \rangle_X = -\frac{2\sqrt{\Gamma}}{\Gamma + 1},$$

whereas dephasing the act record before the filter gives  $\langle \sigma_x \rangle_X = 0$ . Thus the  $\sigma_x$  diagnostic of Sec. VII A is not a new decision-theoretic norm; it tracks off-diagonal coherence in the selected numerator rather than an ordinary diagonal correlation.

The connection with the RQM literature should therefore be read modestly. Di Biagio and Rovelli distinguish relative facts from stable facts, with stability emerging when further interactions can ignore the underlying relativity [21]. Here agreement in the limit  $\Gamma \rightarrow \infty$  is a model analogue of such stability in the computational basis, while finite  $\Gamma$  leaves a measurable mismatch rate.

The construction does not require the stronger cross-perspective-links postulate of Adlam and Rovelli [52], and it does not assume that relative facts are globally consistent across Wigner-friend extensions. Those issues are debated in the current RQM literature [58–60]. For the present model the safe statement is narrower: the parity filter supplies a tunable relation between two records, while the complementarity calculation identifies how much of that relation is diagonal reliability and how much is coherent, basis-dependent visibility.

### D. Simpson's reversal and the postselection flag

The coexistence of a per-attempt one-boxing advantage and conditional-on-success two-boxing dominance is a Simpson reversal [62]. The acceptance event  $A$  depends on both  $x$  and  $y$ , so conditioning on  $A$  is collider conditioning; the size of the reversal is set by the prior. The per-attempt and conditional conventions run parallel to the interventionist and evidential treatments of Newcomb at the level of which probabilities weight the payoffs. The postselection collider is a common effect of  $X$  and  $Y$ , while the textbook setup turns on a common cause (Fig. 2), so we keep the per-attempt and conditional comparisons separate from the causal reading of the CDT/EDT pairing. In Reichenbach's terms [27], the two causal explanations of the correlation are the textbook common cause (a) and the P-CTC direct cause (c); the collider (b) is the excluded case. The CDT/EDT pairing, the per-attempt and conditional conventions, and the reversal itself are one disagreement over which structure governs.

At  $\beta = 1/2$  there is no reversal. The acceptance-rate gradient is  $(p_{\text{match}} - p_{\text{mismatch}})(2\beta - 1)$ , so by Theorem 1(i) the preparation cannot move the acceptance rate there, and the two conventions agree in sign. Away from  $\beta = 1/2$  the preparation does move the acceptance rate, and the success event couples  $X$  and  $Y$  through the survival weights. Conditional scoring then splits by level. Entrywise, for each fixed  $y$  among accepted runs, it removes the survival factor and restores the dominance  $u(D, y) - u(C, y) = m$ , independent of  $\theta$ . At the policy level it integrates over  $Y$  against the posterior  $\Pr(Y \mid A, X = x)$ , which depends on the agent's policy through the same coupling, and the verdicts separate. The separation is widest as  $\beta \rightarrow 1$ : conditional scoring approaches the entrywise two-boxing comparison of  $M$  against  $M + m$ , while per-attempt scoring favors one-boxing across the Stag-Hunt region. The unbiased setup is therefore the symmetric reference case for the Nozick-style stipulation, the reversal recording the effect of departing from it.

### E. Related work

The construction relates to the established quantum game theory program. Beginning with Meyer [63] on quan-



tum penny flip and Eisert, Wilkens and Lewenstein [64] on the quantum Prisoner’s Dilemma, and extended by Piotrowski and Ślaskowski [65] and others into economics, that literature situates quantum effects in the strategy space: classical mixed strategies are replaced with unitary operations on qubits, and new equilibria emerge from the entanglement structure of the initial state, with the Prisoner’s Dilemma rendered trivial in the limit of full entanglement. Piotrowski and Ślaskowski [66] and Chi and Jeong [67] treat Newcomb’s problem itself in this framework, again enlarging the strategy sets to unitary operations. Cavalcanti [68] draws a different quantum connection, reading a Bell scenario as a Newcomb analogue with implications for causal decision theory. The present construction sits at a different point in that design space: strategies remain classical, the agent’s choice of preparation weight, the quantum effect enters through the postselection on the parity sector, and the game-theoretic consequence is a transition from PD to Stag-Hunt ordering at the single threshold  $\Gamma = (M + m)/M$  with the strategy class held fixed.

The accepted-ensemble statistics are those of a pre- and post-selected ensemble [22], and a weak measurement of  $\sigma_x$  before the postselection reads the corresponding weak value [69].

The symmetric game points to a dynamical extension, developed in a companion paper: attractor dynamics in the presence of time machines. The coordination region holds two self-consistent histories, the agreement profiles  $(C, C)$  and  $(D, D)$ , realizing in Newcomb form the multiplicity of self-consistent CTC solutions [70]; in the landscape of Sec. IV B they are its two basins. The deeper well switches at the risk-dominance boundary  $\Gamma = (M + m)/(M - m)$  of Sec. IV; above it  $(C, C)$  is risk-dominant, below it  $(D, D)$  is. Noise in the parity filter drags  $\Gamma_{\text{eff}}$  down, and below the ordering boundary  $\Gamma = (M + m)/M$  the one-boxing basin vanishes.

Lewis’s identification has the agent play against a correlated copy of itself, each reading the other’s act as the forecast of its own [10]. In a time-travel reading  $Y$  may be identified as the agent’s act on a parallel worldline, tied to  $X$  by the consistency loop, so playing against yourself becomes playing against yourself in a parallel worldline.

In the paired-worldline reading, the introduction of mismatches by finite  $\Gamma$  need not violate Novikov’s principle: consistency can be obtained by inverting the reading of the labels  $0 \leftrightarrow 1$ . Let  $r_X(0), r_X(1)$  and  $r_Y(0), r_Y(1)$  denote the physical alternatives encoded by the two binary worldline records. Novikov consistency is the condition

$$r_X(x) = r_Y(y). \quad (52)$$

If the two worldlines use the same coding, Eq. (52) selects raw agreement,  $x = y$ . If one worldline uses the inverted coding, it selects raw opposition,  $x = y \oplus 1$ . Thus both raw classes can describe consistent histories once the worldline label convention is fixed. The role of  $\Gamma$  is to weight these two classes in the selected ensemble.

At fixed preparations, the aligned coding gives unnormalized Born weights  $\alpha_X \alpha_Y$  and  $(1 - \alpha_X)(1 - \alpha_Y)$  for the two raw-agreement histories. The inverted coding gives unnormalized weights  $\alpha_X(1 - \alpha_Y)$  and  $(1 - \alpha_X)\alpha_Y$  for the two raw-opposition histories. In this sense, the system is SWAP-invariant under exchange of the two worldline registers, the point of contact with the single-loop reading of the *Stag Hunt* film.

This analysis therefore extends the reading of P-CTCs by allowing apparent “inconsistent histories” that may be turned consistent across worldline records, if that same consistency is read as a noise within a same world line. It also opens the question of how the multiplicity of consistent world lines may be resolved, and how the present work may relate to quantum gravity, left for future work.

## X. CONCLUSION

This paper replaces Newcomb’s perfect-predictor stipulation by a finite-strength postselected consistency condition. The resulting object is a two-record version of the P-CTC final-boundary map: the selected branch  $K\rho K^\dagger$  followed, when one asks for the accepted state, by the normalized update of Eq. (2). The survival ratio  $\Gamma$  interpolates between no consistency bias and an accepted ensemble dominated by raw label agreement.

The diagonal sector gives the decision and game-theoretic results. At the unbiased setup prior, one-boxing has the larger payoff when the accepted-ensemble distinguishability  $\mathcal{D} = (\Gamma - 1)/(\Gamma + 1)$  exceeds  $m/M$ . There the denominator of Eq. (2) is independent of the agent’s preparation, so the per-attempt and conditional comparisons agree. A classical rejection sampler with the same match and mismatch survival weights reproduces the same diagonal utilities. Under per-attempt scoring the selected branch carries the induced normal form from Prisoner’s-Dilemma to Stag-Hunt ordering at  $\Gamma = (M + m)/M$ , with the one-boxing equilibrium becoming risk-dominant at  $\Gamma = (M + m)/(M - m)$ . Lewis’s side-by-side reading is therefore the weak-predictor end of a one-parameter family.

The off-diagonal sector gives the coherence diagnostic. The accepted state obeys the distinguishability–visibility relation  $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$ , with equality for the coherent parity filter. The residual signal  $\langle \sigma_x \rangle_X$  is absent in a dephased rejection-sampling implementation, so the  $\sigma_x$  expectation separates coherent enforcement from a purely diagonal reliability oracle. Near the perfect-predictor limit this signal scales as  $2/\sqrt{\Gamma}$ , giving an accepted-event scaling of order  $O(\Gamma)$  for fixed precision.

The appendices give the Bell-contraction realization and clarify the scope of the fixed-prior result. Adaptive preparation makes the threshold depend on  $\Omega$ ’s objective. Phase noise replaces  $\Gamma$  by  $\Gamma_{\text{eff}} = \mathbb{E}[\cos^2(\theta/2)]/\mathbb{E}[\sin^2(\theta/2)]$ . The process-matrix appendix records the reduced common-cause embedding of the accepted diagonal slice. The remaining physical question is whether the final-boundary

map of Eq. (2) can be promoted to a spacetime resource.

### ACKNOWLEDGMENTS

The idea of reaching a Stag Hunt through Newcomb's problem and a postselected closed timelike curve was inspired by the short film *Stag Hunt*, directed by Andre Kezzyn. We thank Hippolyte Dourdent for a brief exchange about his process-matrix approach to Newcomb's problem.

### Appendix A: Bell-contraction realization of the filter

The same Kraus operator arises from the postselected-teleportation construction of Lloyd et al. [15]. Let  $L, L'$  be the loop pair prepared and postselected in the Bell state

$$|\Phi\rangle_{LL'} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (\text{A1})$$

For a unitary  $\mathcal{U}_{L,XY}$  acting on the loop qubit  $L$  and the chronology-respecting registers  $XY$ , let  $Z_L := |0\rangle\langle 0|_L - |1\rangle\langle 1|_L$  denote the Pauli  $Z$  on the loop qubit. The successful-branch operator on  $XY$  is the Bell contraction

$$C_{\mathcal{U}} = {}_{LL'}\langle\Phi| (\mathcal{U}_{L,XY} \otimes I_{L'}) |\Phi\rangle_{LL'} = \frac{1}{2} \text{Tr}_L \mathcal{U}_{L,XY}. \quad (\text{A2})$$

Define the parity-dependent angle operator

$$\Theta_\theta = \frac{\theta}{2} \Pi_+ + \left( \frac{\pi}{2} + \frac{\theta}{2} \right) \Pi_-, \quad (\text{A3})$$

and the loop interaction

$$\mathcal{U}_\theta = \exp(iZ_L \otimes \Theta_\theta) = |0\rangle\langle 0|_L \otimes e^{i\Theta_\theta} + |1\rangle\langle 1|_L \otimes e^{-i\Theta_\theta}. \quad (\text{A4})$$

Since  $Z_L$  and  $\Theta_\theta$  are both Hermitian,  $iZ_L \otimes \Theta_\theta$  is anti-Hermitian and  $\mathcal{U}_\theta$  is unitary for every  $\theta$ . Substitution into Eq. (A2) gives

$$\begin{aligned} C_{\mathcal{U}_\theta} &= \frac{1}{2} (e^{i\Theta_\theta} + e^{-i\Theta_\theta}) = \cos \Theta_\theta \\ &= \cos(\theta/2) \Pi_+ + \cos\left(\frac{\pi}{2} + \frac{\theta}{2}\right) \Pi_- \\ &= \cos(\theta/2) \Pi_+ - \sin(\theta/2) \Pi_- = K(\theta). \end{aligned} \quad (\text{A5})$$

Thus the filter is the successful-branch operator of this same construction: one loop qubit with a parity-dependent phase coupling, whose Bell contraction turns the phase difference between agreement and mismatch into the survival amplitudes of Eq. (8). The sharp endpoint is  $\theta = 0$ , where  $K(0) = \Pi_+$  and  $\Gamma = \infty$ . At finite  $\Gamma$  (equivalently  $\theta \neq 0$  in this parametrization), the mismatch sector has selected-branch amplitude  $-\sin(\theta/2)$ . Section VIII treats the different situation where  $\theta$  is not fixed but varies over the ensemble.

### Appendix B: Survival-weight reduction

This appendix proves the proposition of Sec. II E. By diagonality,  $K$  acts as scalar multiplication on each computational basis state. Joint bit-flip exchanges  $|00\rangle \leftrightarrow |11\rangle$  and  $|01\rangle \leftrightarrow |10\rangle$ , forcing  $K_{00} = K_{11}$  and  $K_{01} = K_{10}$ . Setting  $k_+ := K_{00} = K_{11}$  and  $k_- := K_{01} = K_{10}$  gives

$$K = k_+ \Pi_+ + k_- \Pi_-.$$

The effect  $E = K^\dagger K$  is diagonal in the same basis with diagonal entries  $|K_{xy}|^2$ . For basis-diagonal input

$$\rho(\alpha, \beta) = \sum_{xy} \rho_{xy}(\alpha, \beta) |xy\rangle \langle xy|$$

and basis-diagonal payoff

$$U = \sum_{xy} u(x, y) |xy\rangle \langle xy|,$$

the per-attempt expectation is

$$\langle U \rangle_{\text{per-att}}(\alpha) = \sum_{xy} \rho_{xy}(\alpha, \beta) p_{xy} u(x, y).$$

The policy difference between two preparations is linear in  $(p_{\text{match}}, p_{\text{mismatch}})$  and homogeneous of degree one, so its sign depends only on the ratio  $\Gamma = p_{\text{match}}/p_{\text{mismatch}}$ .

### Appendix C: Proof of Theorem 1

This appendix proves Theorem 1. Part (i) is the displayed differentiation. For (ii), at  $\beta = 1/2$  the per-attempt payoffs are  $\bar{u}(C) = \frac{1}{2} p_{\text{match}} M$  and  $\bar{u}(D) = \frac{1}{2} [p_{\text{mismatch}}(M+m) + p_{\text{match}} m]$ , so  $\bar{u}(C) > \bar{u}(D)$  iff  $p_{\text{match}}(M-m) > p_{\text{mismatch}}(M+m)$  iff  $\Gamma > (M+m)/(M-m)$ . The conditional comparison is Eq. (D5),  $U_A(D) - U_A(C) = [(M+m) - \Gamma(M-m)]/(\Gamma+1)$ , which is negative on the same set. Part (iii) substitutes  $q = \Gamma/(\Gamma+1)$  into the threshold (Sec. III D).  $\square$

### Appendix D: Utilities conditioned on success

An alternative convention evaluates utilities conditioned on postselection success: one conditions on A and then computes expected payoff. In that convention, the factor  $p_{xy}$  in Eq. (16) is renormalized away. The main text keeps  $p_{xy}$  visible by working per attempt. Concretely, let  $\mathbb{P}(x, y)$  denote the joint distribution over profiles before the ancilla is measured. Conditioning on A produces a new distribution over the accepted runs,

$$\mathbb{P}(x, y | A) = \frac{p_{xy} \mathbb{P}(x, y)}{\sum_{x', y'} p_{x'y'} \mathbb{P}(x', y')}, \quad (\text{D1})$$

and the conditional expected utility is

$$\mathbb{E}[u_i | A] = \sum_{x, y} \mathbb{P}(x, y | A) u_i(x, y). \quad (\text{D2})$$

For the Newcomb payoffs (17), conditional scoring restores dominance entrywise:

$$\begin{aligned} u_1(D, C) - u_1(C, C) &= (M + m) - M = m, \\ u_1(D, D) - u_1(C, D) &= m - 0 = m. \end{aligned} \quad (\text{D3})$$

Thus  $D$  strictly dominates  $C$  in the conditional-on-success normal form, independently of  $\theta$ . Per-attempt valuation keeps the survival factors  $p_{xy}$  attached to the payoffs; conditional-on-success valuation renormalizes them away.

The entrywise dominance just computed holds with the prior-record variable  $Y$  held fixed. The policy-level conditional expected utility  $U_A(x) := E[u_1 \mid A, X = x]$  integrates over  $Y$  against  $\Pr(Y \mid A, X = x)$ , and the posterior depends on  $x$  because  $A$  couples  $X$  and  $Y$  through the success weights:

$$\Pr(Y = y \mid A, X = x) = \frac{p_{xy} \Pr(Y = y)}{\sum_{y'} p_{xy'} \Pr(Y = y')}. \quad (\text{D4})$$

For the Newcomb payoffs at the unbiased setup prior  $\beta = 1/2$ ,  $U_A(C) = M\Gamma/(\Gamma + 1)$  and  $U_A(D) = (M + m + m\Gamma)/(\Gamma + 1)$ , so

$$U_A(D) - U_A(C) = \frac{(M + m) - \Gamma(M - m)}{\Gamma + 1}, \quad (\text{D5})$$

and the policy comparison favors one-boxing for  $\Gamma > (M + m)/(M - m)$ , the risk-dominance boundary of Sec. IV.

For general  $\beta \in (0, 1)$ ,

$$\begin{aligned} U_A(C) &= \frac{\beta\Gamma M}{\beta\Gamma + 1 - \beta}, \\ U_A(D) &= \frac{\beta(M + m) + (1 - \beta)\Gamma m}{\beta + (1 - \beta)\Gamma}. \end{aligned} \quad (\text{D6})$$

Thus  $U_A(C) > U_A(D)$  iff

$$\begin{aligned} \beta(1 - \beta)(M - m)\Gamma^2 + m(2\beta(1 - \beta) - 1)\Gamma \\ - \beta(1 - \beta)(M + m) > 0. \end{aligned} \quad (\text{D7})$$

At  $\beta = 1/2$  this reduces to  $\Gamma > (M + m)/(M - m)$ , the risk-dominance boundary. As  $\beta \rightarrow 1$ , the comparison tends to the fixed-record entrywise result  $M$  versus  $M + m$ , so conditional scoring favors two-boxing.

The entrywise statement is therefore a feature of fixed- $Y$  accepted branches; the policy comparison that the agent actually faces is prior-dependent.

## Appendix E: Payoff-ordering classification

This appendix states and proves the payoff-ordering proposition of Sec. IV, with entries (27).

**Proposition** (Structure and payoff-ordering boundaries). *Let  $M > m > 0$ , let  $0 < p_{\text{match}}, p_{\text{mismatch}} < 1$ , define  $\Gamma := p_{\text{match}}/p_{\text{mismatch}}$ , and score utilities per attempt. The postselected agreement filter induces the symmetric normal form with  $R = p_{\text{match}}M$ ,  $T = p_{\text{mismatch}}(M + m)$ ,  $P = p_{\text{match}}m$ , and  $S = 0$ . Then:*

- (i) *The symmetrized table has strict Prisoner's-Dilemma ordering,  $T > R > P > S$ , iff  $\Gamma < (M + m)/M$ .*
- (ii) *The symmetrized table has the Stag-Hunt/assurance coordination structure,  $R > T$ ,  $R > P$ , and  $P > S$ , iff  $\Gamma > (M + m)/M$ .*
- (iii) *Within this coordination region, the strict ordinal Stag-Hunt chain  $R > T > P > S$  holds iff  $(M + m)/M < \Gamma < (M + m)/m$ .*
- (iv) *For  $\Gamma > (M + m)/m$ , the ordinal ordering is  $R > P > T > S$ . This is a strong-assurance, or diagonal-dominant, coordination regime: the fixed-point structure is unchanged, but mutual defection exceeds the unilateral temptation payoff per attempt.*
- (v) *At  $\Gamma = (M + m)/M$  one has  $R = T$ , so  $D$  weakly dominates  $C$  and  $(C, C)$  is a weak fixed point. At  $\Gamma = (M + m)/m$  one has  $T = P$ , so the ordinal chain degenerates to  $R > P = T > S$ ; the temptation payoff per attempt has fallen to the level of mutual defection.*
- (vi) *Within the coordination region,  $(C, C)$  is risk-dominant iff  $R - T > P - S$ , equivalently  $\Gamma > (M + m)/(M - m)$ .*

*In the limiting endpoint cases, the normal form degenerates further:  $R = P = S = 0$  when  $p_{\text{match}} = 0$ , and  $T = S = 0$  when  $p_{\text{mismatch}} = 0$ .*

*Proof.* Since  $M > m > 0$  and  $p_{\text{match}}, p_{\text{mismatch}} > 0$ , we have  $R > P > S$  and  $T > S$ . The comparison between  $R$  and  $T$  gives  $R > T$  iff  $p_{\text{match}}M > p_{\text{mismatch}}(M + m)$ , iff  $\Gamma > (M + m)/M$ . This is the boundary between Prisoner's-Dilemma ordering and the Stag-Hunt/assurance coordination structure. The remaining ordinal comparison is  $T > P$ , which holds iff  $p_{\text{mismatch}}(M + m) > p_{\text{match}}m$ , iff  $\Gamma < (M + m)/m$ . Combining these inequalities gives the strict ordinal Stag-Hunt chain  $R > T > P > S$ ; reversing the second gives  $R > P > T > S$ . Finally,  $R - T > P - S$  iff  $p_{\text{match}}M - p_{\text{mismatch}}(M + m) > p_{\text{match}}m$ , iff  $p_{\text{match}}(M - m) > p_{\text{mismatch}}(M + m)$ , iff  $\Gamma > (M + m)/(M - m)$ .  $\square$

## Appendix F: Potential landscape and basins

This appendix describes the potential, basins, and barrier for the symmetric two-player reading in which  $Y$  is a second decision register; in the single-agent reading they are the structure the setup prior  $\beta$  inherits. Writing  $x \in [0, 1]$  for the one-boxing weight of the symmetric profile, the selection gradient  $U(C \mid x) - U(D \mid x) = \Delta x - (P - S)$  integrates to the potential

$$\Phi(x) = \frac{1}{2} \Delta x^2 - (P - S)x, \quad \Delta := R - S - T + P, \quad (\text{F1})$$

which under replicator dynamics is climbed according to  $\dot{\Phi} = x(1-x)[U(C|x) - U(D|x)]^2 \geq 0$ , so  $\Phi$  is a Lyapunov function for that flow. In the coordination region  $\Delta > 0$  the two pure profiles  $x = 0$  (two-boxing) and  $x = 1$  (one-boxing) are the attractors, separated by the interior repeller  $x^* = (P - S)/\Delta = \beta_{\text{ind}}$  of Eq. (30). Reading  $V := -\Phi$  as a potential landscape gives a double well whose minima are the two equilibria and whose barrier sits at  $\beta_{\text{ind}}$ ; with the Newcomb entries (27) the barrier height above the two-boxing well is

$$V(x^*) - V(0) = \frac{P^2}{2\Delta} = \frac{p_{\text{match}}^2 m^2}{2(M+m)(p_{\text{match}} - p_{\text{mismatch}})}, \quad (\text{F2})$$

fixed in shape by  $\Gamma$  and  $m/M$ . The one-boxing equilibrium is the global maximizer of  $\Phi$  (equivalently the deeper well of  $V$ ) iff  $\Phi(1) > \Phi(0)$ , i.e.  $\Delta > 2P$ , i.e.  $R - T > P - S$ : the risk-dominance condition (31),  $\Gamma > (M + m)/(M - m)$ . The risk-dominance boundary is therefore the value of  $\Gamma$  at which the global minimum of the landscape switches wells.

*First-order structure.* Equation (F1) is a Landau potential in the order parameter  $x$ , the one-boxing weight, with  $\Delta = (M + m)(p_{\text{match}} - p_{\text{mismatch}})$  and  $P - S = p_{\text{match}}m$ . The interior extremum  $x^* = (P - S)/\Delta$  enters  $[0, 1]$  at  $\Gamma = (M + m)/M$ : below it the potential is monotone, with the single two-boxing well  $x = 0$ ; at that value the one-boxing well  $x = 1$  and the barrier appear together. This is the metastability onset, the spinodal of the one-boxing phase, coinciding with the Prisoner's-Dilemma-to-Stag-Hunt boundary.

The well-depth difference is

$$\Phi(1) - \Phi(0) = \frac{1}{2} [p_{\text{match}}(M - m) - p_{\text{mismatch}}(M + m)], \quad (\text{F3})$$

negative through the metastable interval and zero at  $\Gamma = (M + m)/(M - m)$ , where the wells are degenerate. This is the first-order transition: the deeper well switches from  $x = 0$  to  $x = 1$ , the equilibrium order parameter jumping discontinuously, with the risk-dominance boundary as the coexistence point. The two-boxing well persists above the transition, its barrier approaching  $x^* \rightarrow m/(M + m)$  as  $\Gamma \rightarrow \infty$ , so the transition is one-sided, with a single spinodal at finite  $\Gamma$ . Regions I, II, and III of Fig. 3 are the single-well, metastable, and switched phases.

## Appendix G: Side-by-side decomposition

This appendix states and proves the side-by-side proposition of Sec. V.

**Proposition (side-by-side decomposition and the PD→SH transition).** *Let  $M > m > 0$  and score utilities per attempt.*

- (i) *For every  $\Gamma \geq 1$ , read parametrically, each player's decision in  $G(\Gamma)$  is the standard Newcomb problem with payoffs  $u_1$  and predictor reliability  $q = \Gamma/(\Gamma +$*

*1): two-boxing dominates entrywise,  $u_1(D, y) - u_1(C, y) = m > 0$ , while one-boxing carries the larger expected utility iff  $q > (M + m)/2M$ , i.e.  $\Gamma > (M + m)/(M - m)$ .*

- (ii) *Read as a game,  $G(\Gamma)$  is Prisoner's-Dilemma-ordered for  $\Gamma < (M + m)/M$  and Stag-Hunt-ordered for  $\Gamma > (M + m)/M$ , with  $(C, C)$  risk-dominant iff  $\Gamma > (M + m)/(M - m)$  (Prop. E).*
- (iii) *The thresholds of (i) and (ii) coincide: each side's Newcomb verdict turns to one-boxing at the same  $\Gamma = (M + m)/(M - m)$  at which  $(C, C)$  becomes risk-dominant in the composite.*

Hence  $G(\Gamma)$  is two Newcomb problems side by side, in Lewis's sense, throughout the predictor regime  $\Gamma \geq 1$ ; the consistency factor changes only the equilibrium structure of the game they compose into. The crossover at  $\Gamma = (M + m)/M$  is a transition of a genuine two-player game built from two Newcomb components, carrying it from Prisoner's-Dilemma to Stag-Hunt ordering.

*Proof.* (i) Holding the prediction  $x_j$  fixed, the raw comparison is  $u_1(D, y) - u_1(C, y) = m > 0$ . With reliability  $q$ ,  $V_{\text{one}}(q) = Mq$  and  $V_{\text{two}}(q) = M + m - Mq$  (Sec. IIID), so one-boxing wins iff  $q > (M + m)/2M$ ; substituting  $q = \Gamma/(\Gamma + 1)$  gives  $\Gamma > (M + m)/(M - m)$ . (ii) is Prop. E. (iii) equates the two thresholds.  $\square$

## Appendix H: Forecasting-predictor optimum

With  $W$  as defined in Eq. (38), under the best response (40), the agent's reduced payoff is  $W(\alpha, 1) = \alpha R + (1 - \alpha)T$  for  $\alpha > \frac{1}{2}$  and  $W(\alpha, 0) = \alpha S + (1 - \alpha)P$  for  $\alpha < \frac{1}{2}$ , each affine in  $\alpha$ . The lower branch is decreasing (since  $P > S$ ), with maximum  $P$  at  $\alpha = 0$ . The upper branch has slope  $R - T$ . In the Stag-Hunt region  $R > T$  it is increasing, so its maximum is  $R$  at  $\alpha = 1$ ; because  $R > P$  always, the global optimum is the pure corner  $\alpha_{\text{MAP}}^* = 1$ , Eq. (41).

In the Prisoner's-Dilemma region  $R < T$  the upper branch  $W(\alpha, 1)$  decreases in  $\alpha$ , so the agent prefers  $\alpha$  just above  $\frac{1}{2}$ . The payoff approaches  $(R + T)/2$  as  $\alpha \rightarrow \frac{1}{2}^+$  but no preparation attains it, since at  $\alpha = \frac{1}{2}$  the maximum-a-posteriori tie-break flips  $\beta$  to the opposite corner; optimizing against this threshold discontinuity yields only a supremum. The coordination region is clean for the complementary reason: there one-boxing and being forecast as a one-boxer align, so the optimum is the pure corner  $\alpha = 1$ .

## Appendix I: Self-calibrated predictor

A calibrated predictor matched to the realized accepted-run frequency sets  $\beta = \Pr(X=C | A)$ . With  $\delta := p_{\text{match}} -$



$p_{\text{mismatch}}$ ,  $w_C(\beta) = p_{\text{mismatch}} + \delta\beta$ ,  $w_D(\beta) = p_{\text{match}} - \delta\beta$ , the condition is linear in  $\alpha$  and inverts as

$$\alpha(\beta) = \frac{\beta w_D}{Z}, \quad 1 - \alpha(\beta) = \frac{(1 - \beta) w_C}{Z},$$

$$Z = p_{\text{mismatch}} + 2\delta\beta(1 - \beta). \quad (\text{I1})$$

Since  $d\alpha/d\beta = p_{\text{mismatch}} [p_{\text{match}} - 2\delta\beta(1 - \beta)]/Z^2 > 0$ , the map is an increasing bijection of  $[0, 1]$  and the fixed point is unique, with  $\alpha = 1 \Leftrightarrow \beta = 1$ . Along this curve the per-attempt payoff (38) is

$$W(\beta) = \frac{\beta^2 w_D R + \beta(1 - \beta) w_C T + (1 - \beta)^2 w_C P}{Z}, \quad (\text{I2})$$

and a direct computation gives

$$R - W(\beta) = \frac{(1 - \beta) Q(\beta)}{Z}, \quad (\text{I3})$$

with  $Q$  quadratic, leading coefficient  $\delta [m(p_{\text{match}} - p_{\text{mismatch}}) - M(p_{\text{match}} + p_{\text{mismatch}})] \leq 0$  for  $\Gamma \geq 1$ , so  $Q$  is concave on  $[0, 1]$ . Its endpoints,  $Q(0) = p_{\text{match}} p_{\text{mismatch}} (M - m) > 0$  and  $Q(1) = p_{\text{match}} (M p_{\text{match}} - m p_{\text{mismatch}}) > 0$ , are positive; concavity then gives  $Q > 0$  throughout, so  $R - W(\beta) \geq 0$  with equality iff  $\beta = 1$ . Hence the agent's optimum is  $\alpha^* = 1$  with payoff  $R$  for every  $\Gamma \geq 1$ , matching the verdict under the  $\beta = \alpha$  predictor. The two notions of calibration differ only in the off-optimum response.

## Appendix J: Adversarial adaptive preparation

This appendix gives the lower-envelope calculation quoted in Sec. VI. Here  $W$  is the payoff in Eq. (38) and  $\Delta = R - S - T + P$  as in Eq. (39). The adversarial objective lets the predictor choose the setup prior  $\beta \in [0, 1]$  after observing the Born weight  $\alpha$ , minimizing the agent's per-attempt payoff:

$$V_{\text{adv}}(\alpha) := \inf_{\beta \in [0, 1]} W(\alpha, \beta).$$

Since  $W$  is affine in  $\beta$ , with

$$\frac{\partial W}{\partial \beta} = T - P + \alpha \Delta,$$

the minimizing prior lies at one of the two corners of the interval. In the predictor regime  $\Gamma \geq 1$ , the Newcomb entries give  $\Delta \geq 0$ . If  $T \geq P$ , then  $\partial W/\partial \beta \geq 0$  for every  $\alpha$ , so the minimizing prior is  $\beta = 0$ . Hence

$$V_{\text{adv}}(\alpha) = W(\alpha, 0) = \alpha S + (1 - \alpha)P.$$

For the Newcomb matrix (27),  $P > S$ , so this lower envelope is maximized at  $\alpha = 0$ . The robust adaptive preparation is therefore pure two-boxing whenever  $T \geq P$ . At  $\Gamma = 1$ , the filter has no consistency bias,  $\Delta = 0$ , and  $T - P = p_{\text{match}} M > 0$ , so the same conclusion holds.

If  $T < P$ , the lower envelope has an interior kink, determined by

$$W(\alpha, 0) = W(\alpha, 1).$$

Using Eq. (39), this gives

$$\alpha^* = \frac{P - T}{\Delta}. \quad (\text{J1})$$

The dual prior that makes the agent indifferent satisfies

$$\frac{\partial W}{\partial \alpha} = 0,$$

hence

$$\beta^* = \frac{P - S}{\Delta}. \quad (\text{J2})$$

For the Newcomb entries (27), the condition  $T < P$  is

$$p_{\text{mismatch}}(M + m) < p_{\text{match}} m, \quad \Gamma > \frac{M + m}{m}. \quad (\text{J3})$$

Substituting the Newcomb entries into Eqs. (J1)–(J2) gives

$$\alpha^* = \frac{P - T}{\Delta} = \frac{\Gamma m - (M + m)}{(M + m)(\Gamma - 1)}, \quad (\text{J4})$$

$$\beta^* = \frac{P - S}{\Delta} = \frac{\Gamma m}{(M + m)(\Gamma - 1)}. \quad (\text{J5})$$

The setup prior  $\beta^*$  coincides with the fixed- $\beta$  basin boundary  $\beta_{\text{ind}}$  in Eq. (30); the quantity  $\alpha^*$  is the Born weight of one-boxing in the lower-envelope adaptive preparation. The two quantities are generally distinct:

$$\beta^* - \alpha^* = \frac{1}{\Gamma - 1}.$$

This gap closes as  $\Gamma$  grows, and at  $\Gamma \rightarrow \infty$  both reach

$$\frac{m}{M + m}.$$

Thus the adversarial robust preparation remains overwhelmingly two-boxing in the standard Newcomb regime  $M \gg m$ , even in the ideal consistency limit.

## Appendix K: Complementarity and the $\sigma_x$ signature

This appendix proves the distinguishability–visibility bound of Sec. VII and its corollary, and gives the symmetric-preparation state algebra behind  $\langle \sigma_x \rangle_X$ .

*Proof of the bound (46).* Read the computational basis of  $X$  as the two alternatives in the standard two-path comparison. The register  $Y$  is a record whose readout gives a candidate guess of the later  $X$ -basis outcome. In Englert's formulation, the distinguishability entering the duality relation is the best which-way information obtainable from the record; Jaeger–Shimony–Vaidman give the

corresponding two-alternative visibility tradeoff [33, 34]. The fixed record readout used in the Newcomb payoff calculation has bias  $\mathcal{D} = 2q - 1$ , which is at most that optimized distinguishability. Applying the standard bound to the optimized quantity therefore also bounds this  $\mathcal{D}$ :

$$\mathcal{D}^2 + \mathcal{V}^2 \leq 1.$$

Equality is not assumed by the bound; it is verified below for the coherent real-phase parity filter, where the computational-basis readout of  $Y$  is already optimal and the accepted state is pure.  $\square$

*Proof of the no-early-fact corollary.*  $\Pr(y = x) = 1$  with a designated computational-basis readout of  $Y$  means the joint distribution of the  $X$ -readout and the  $Y$ -readout is supported on the diagonal. The joint state may still carry coherences inside that diagonal subspace, but it has the form

$$\rho_{XY} = \sum_{y, y'} r_{yy'} |y\rangle_X |y'\rangle_Y \langle y'|_X \langle y|_Y$$

for some matrix  $r_{yy'}$ . Tracing over  $Y$  removes the  $y \neq y'$  terms, so

$$\rho_X = \sum_y r_{yy} |y\rangle \langle y|$$

is diagonal and  $\mathcal{V} = 0$ . A perfect classical readout of  $X$  is created by a  $\{C, D\}$ -basis measurement, which leaves the diagonal Born weights untouched and removes only the off-diagonal coherence, so  $p_C = \alpha$ . The endpoint  $\Gamma \rightarrow \infty$  of (47) realizes this, with  $\mathcal{D} \rightarrow 1$ ,  $\mathcal{V} \rightarrow 0$ .  $\square$

*The  $\sigma_x$  state algebra.* The two CNOTs from  $X$  and  $Y$  into the ancilla map  $|+\rangle_X \otimes |+\rangle_Y \otimes |0\rangle_a$  to

$$\frac{1}{2}(|00\rangle|0\rangle + |01\rangle|1\rangle + |10\rangle|1\rangle + |11\rangle|0\rangle)_{XYa}, \quad (\text{K1})$$

with the ancilla carrying the parity  $x \oplus y$ . The rotation  $R_y(\theta)$  on the ancilla followed by postselection on  $|0\rangle_a$  yields the normalized joint state

$$|\psi(\theta)\rangle_{XY} = \frac{1}{\sqrt{2}}(c|00\rangle - s|01\rangle - s|10\rangle + c|11\rangle), \quad (\text{K2})$$

with  $c = \cos(\theta/2)$  and  $s = \sin(\theta/2)$ . At the symmetric preparation each parity sector has prior weight  $1/2$ , so the acceptance probability is  $(c^2 + s^2)/2 = 1/2$ , independent of  $\theta$ . Within the accepted ensemble the match and mismatch probabilities are  $c^2$  and  $s^2$ , recovering  $\Gamma = \cot^2(\theta/2)$ .

The reduced state on  $X$  is

$$\rho_X(\theta) = \frac{1}{2}(I - \sin \theta \cdot \sigma_x) = \frac{1}{2} \left( I - \frac{2\sqrt{\Gamma}}{\Gamma + 1} \sigma_x \right), \quad (\text{K3})$$

so  $\langle \sigma_x \rangle_X = -2\sqrt{\Gamma}/(\Gamma + 1)$ .

The distinguishability comes from the same state. Grouped by the act value,  $|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}(|C\rangle_X |\phi_C\rangle_Y + |D\rangle_X |\phi_D\rangle_Y)$  with conditional records  $|\phi_C\rangle = c|0\rangle - s|1\rangle$

and  $|\phi_D\rangle = -s|0\rangle + c|1\rangle$ . Their density-matrix difference is diagonal,

$$|\phi_C\rangle\langle\phi_C| - |\phi_D\rangle\langle\phi_D| = (c^2 - s^2)\sigma_z = \cos \theta \sigma_z, \quad (\text{K4})$$

so, in the two-path language, the computational-basis readout of  $Y$  already attains the optimal which-way measurement, with  $\mathcal{D} = \frac{1}{2}\|\cos \theta \sigma_z\|_1 = \cos \theta$ . With  $\mathcal{V} = \sin \theta$  from Eq. (K3), the coherent filter saturates the bound,  $\mathcal{D}^2 + \mathcal{V}^2 = \cos^2 \theta + \sin^2 \theta = 1$ , recovering Eq. (47). This is the equality case deferred in the proof of (46): the accepted state is pure and its record measurement optimal.

The local signature depends on the implementation phase: for the symmetric phase family  $K = c\Pi_+ + e^{i\varphi}s\Pi_-$  the local  $\sigma_x$  signal is proportional to  $\cos \varphi$ , vanishing at  $\varphi = \pi/2$  with no compensating  $\sigma_y$  signal, so the real-phase  $R_y(\theta)$  circuit maximizes it. The same scaling is obtained from the two-qubit postselection benchmark: prepare both registers in  $|+\rangle$ , apply the parity-filter circuit of Fig. 1, condition on the ancilla outcome  $|0\rangle$ , and evaluate  $\sigma_x$  on the act register. Let

$$s_\Gamma := |\langle \sigma_x \rangle_X| = \frac{2\sqrt{\Gamma}}{\Gamma + 1}.$$

For a selected-branch estimate of  $\langle \sigma_x \rangle_X$ , bounded variance gives an accepted-event count of order the inverse squared signal,

$$N_{\text{acc}} \gtrsim \frac{c}{s_\Gamma^2} = \frac{c(\Gamma + 1)^2}{4\Gamma} = O(\Gamma),$$

where  $c$  is a confidence-dependent constant. For the symmetric coherent input the success probability is  $1/2$ , so the preselection count differs only by this constant factor. The diagnostic becomes smallest where prediction is sharpest and the act register is being dephased.

## Appendix L: Maximal-ignorance limit

The maximally noisy case of Sec. VIII is  $\theta$  uniform on  $[0, \pi]$ . The angle is the natural ignorance variable: it parameterizes the rotation gate directly, and the uniform prior is symmetric about  $\theta = \pi/2$ , so the averaged survival weights coincide; a flat prior in  $\Gamma$  or in  $p_{\text{mismatch}}$  would privilege one sector by construction. At fixed  $\theta$ , the filter is Stag-Hunt-ordered for  $\theta < \theta^* \equiv 2 \arctan \sqrt{M/(M+m)}$ , so  $\Pr(\text{SH}) = \theta^*/\pi$  and  $\Pr(\text{PD}) - \Pr(\text{SH}) = 1 - 2\theta^*/\pi = m/(\pi M) + O(m^2/M^2)$ : the hidden split is close to even, with a small Prisoner's-Dilemma excess set by the transparent-box payoff.

The ordering of the averaged channel is not the run-by-run majority, because the payoff ordering is not linear in the sector weights. For  $\theta$  uniform,  $\mathbb{E}[\cos^2(\theta/2)] = \mathbb{E}[\sin^2(\theta/2)] = \frac{1}{2}$ , so  $\Gamma_{\text{eff}} = 1$ : the averaged game is Prisoner's-Dilemma-ordered, even though half the fixed- $\theta$  values lie on the Stag-Hunt side. At  $\Gamma_{\text{eff}} = 1$  agreement and disagreement survive equally, so the filter carries no

match-mismatch signal. The averaged game is Stag-Hunt-ordered only when  $\mathbb{E}[\cos^2(\theta/2)] > (M+m)/(2M+m)$ . The agent, lacking the realized  $\theta$ , sees only  $E_{\text{eff}}$ , and its payoff is governed by  $\Gamma_{\text{eff}}$ .

### Appendix M: Process-matrix form and causal separability

Process matrices were introduced by Oreshkov, Costa and Brukner as a framework for assigning probabilities to local quantum operations without assuming a fixed global causal order; some valid process matrices violate causal inequalities [49]. Later work sharpened the distinction between causal correlations and causal separability [50]. Araújo, Guérin and Baumeler also showed that process matrices correspond to a linear special case of postselected CTC computation [48]. Here the formalism is used only for the accepted slice of the Newcomb filter. A full process-matrix realization of the heralded Newcomb filter would instead have to represent the pre-conditioning probabilities  $P(x, y, A)$ , specify the local input and output systems and the heralding event, and verify the process-validity constraints for arbitrary local instruments. The AF/BW-Lugano reference is included because it is the standard classical, diagonal, consistency-based process without a predefined causal order [35, 36]. Those features make it the closest process-matrix comparison to a full Newcomb-style agreement constraint. The appendix below has the narrower role of recording the reduced common-cause slice used in the payoff sections and showing that the accepted diagonal statistics do not require causal nonseparability.

For the present filter, the trace-decreasing numerator  $\rho \mapsto K\rho K^\dagger$  is linear, while the normalized postselected update  $\rho \mapsto K\rho K^\dagger / \text{Tr}(K\rho K^\dagger)$  is linear only on preparation families for which the success probability is constant. By Theorem 1(i), this constancy holds on the diagonal preparation family used for the payoff comparison when  $\beta = 1/2$ . The process matrix below implements the narrower check described above: it embeds the accepted diagonal slice after success has been conditioned on. The calculation shows that this payoff-level correlation, considered as a diagonal distribution, has a causally separable bipartite embedding.

The uniform-marginal member of this slice is the symmetric preparation  $\alpha = \beta = 1/2$ . Before conditioning, the four branches  $(x, y)$  have equal weight; the heralding event then distinguishes only the agreement and disagreement sectors. The accepted ensemble therefore has total agreement probability

$$q := \Pr(X = Y | A) = \frac{p_{\text{match}}}{p_{\text{match}} + p_{\text{mismatch}}} = \frac{\Gamma}{\Gamma + 1},$$

$$\mathcal{D} = 2q - 1 = \frac{\Gamma - 1}{\Gamma + 1}.$$
(M1)

The two agreement branches share weight  $q$ , and the two disagreement branches share weight  $1 - q$ . With  $z_C = +1$

and  $z_D = -1$ , this four-cell distribution is equivalently

$$\Pr(x, y | A) = \frac{1}{4}(1 + \mathcal{D}z_x z_y), \quad z_C = +1, \quad z_D = -1. \quad (\text{M2})$$

The absence of terms proportional to  $z_x$  or  $z_y$  records the symmetric preparation: after conditioning, both one-register marginals remain uniform, and the only diagonal statistic is the record-act correlation. For  $\beta = 1/2$  but  $\alpha \neq 1/2$ , the accepted distribution is still diagonal and causally separable, but its one-register marginals are not uniform; Eq. (M2) is the symmetric point used to test whether the Newcomb correlation itself carries causal-order structure.

Embed Eq. (M2) as a bipartite process with qubit input spaces  $A_I, B_I$  and unused qubit output spaces  $A_O, B_O$ :

$$W = \frac{1}{4}(I + \mathcal{D}\sigma_z^{A_I} \otimes \sigma_z^{B_I}) \otimes I^{A_O B_O}, \quad \mathcal{D} = \frac{\Gamma - 1}{\Gamma + 1}. \quad (\text{M3})$$

In this reduced embedding the output factors are spectrators, included to use the standard process-matrix normalization. Eq. (M3) stores the accepted record distribution. For computational-basis readouts take the local instrument elements

$$M_A^x = \Pi_x^{A_I} \otimes \frac{I^{A_O}}{2}, \quad M_B^y = \Pi_y^{B_I} \otimes \frac{I^{B_O}}{2}, \quad (\text{M4})$$

with  $\Pi_x = (I + z_x \sigma_z)/2$ . The process-matrix Born rule gives

$$\text{Tr}[W(M_A^x \otimes M_B^y)] = \frac{1}{4}(1 + \mathcal{D}z_x z_y), \quad (\text{M5})$$

and therefore

$$\Pr(X = Y) = \sum_x \text{Tr}[W(M_A^x \otimes M_B^x)] = \frac{1 + \mathcal{D}}{2} = \frac{\Gamma}{\Gamma + 1} = q. \quad (\text{M6})$$

Equation (M3) is a valid process matrix. Positivity follows from  $|\mathcal{D}| \leq 1$ . The trace is

$$\text{Tr} W = d_{A_O} d_{B_O} = 4, \quad (\text{M7})$$

and the valid-subspace condition is immediate because  $W$  is proportional to the identity on both output spaces:

$$W = {}_{A_O} W = {}_{B_O} W = {}_{A_O B_O} W, \quad {}_X W := \frac{1}{d_X} I_X \otimes \text{Tr}_X W. \quad (\text{M8})$$

Equivalently, it satisfies the standard bipartite condition  $W = {}_{A_O} W + {}_{B_O} W - {}_{A_O B_O} W$ .

The reduced process is therefore causally separable: it is a common-cause embedding of the accepted diagonal statistics, with no remaining dependence on either party's output. This statement concerns the post-heralding slice. The physical route producing that slice is the trace-decreasing filter itself: in the laboratory reading the association is generated by conditioning on  $A$ , while in the cyclic P-CTC reading the same structure is read as final-boundary influence on the earlier forecast register. For the reduced slice, no local operation by  $A$  can signal to  $B$ , and no local operation by  $B$  can signal to  $A$ . It lies in

the intersection of the  $A \prec B$  and  $B \prec A$  process cones, so one may write a separable decomposition, for example

$$W = \frac{1}{2}W^{A \prec B} + \frac{1}{2}W^{B \prec A}, \quad W^{A \prec B} = W^{B \prec A} = W. \quad (\text{M9})$$

The Pauli support on the input spaces is only  $I \otimes I$  and  $\sigma_z \otimes \sigma_z$ , the diagonal correlation of a common cause. Within this reduced embedding, the Newcomb payoff comparison uses only that common-cause record correlation.

This also separates the two notions of “quantum” used in the paper. The process matrix in Eq. (M3) captures the accepted computational-basis statistics that enter the payoff comparison. The  $\sigma_x$  diagnostic of Sec. VII A belongs instead to the coherent numerator  $K\rho K^\dagger$ : it is sensitive to coherent parity-sector terms before the

diagonal readout. Causal nonseparability is a different diagnostic.

Finally, Eq. (M3) is compatible with the recent parity-erasure characterization of valid process matrices [72]. The parity erased there is an input–output signaling parity in local process tomography, whereas the record parity  $x \oplus y$  of Sec. II is the agreement variable selected by the heralding event. In the reduced  $W$  above, the output systems are spectators, so the input–output parity is absent; the record parity remains as an ordinary input–input correlation in the accepted records. The reduced  $W$  displayed here is the post-heralding diagnostic; the heralding process itself is represented in the trace-decreasing map.

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- [1] R. Nozick, “Newcomb’s problem and two principles of choice,” in *Essays in Honor of Carl G. Hempel*, edited by N. Rescher (D. Reidel, 1969), pp. 114–146.
  - [2] M. Gardner, “Free will revisited, with a mind-bending prediction paradox by William Newcomb,” *Mathematical Games*, *Scientific American* **229**(1), 104 (July 1973).
  - [3] M. Gardner, “Reflections on Newcomb’s problem,” *Mathematical Games*, *Scientific American* **230**(3), 102 (March 1974).
  - [4] A. Ahmed, *Evidence, Decision and Causality* (Cambridge University Press, 2014).
  - [5] J. L. Mackie, “Newcomb’s paradox and the direction of causation,” *Canadian Journal of Philosophy* **7**(2), 213–225 (1977).
  - [6] P. Weirich, “Causal Decision Theory,” in *The Stanford Encyclopedia of Philosophy* (Winter 2024 edition), edited by E. N. Zalta and U. Nodelman. <https://plato.stanford.edu/archives/win2024/entries/decision-causal/>.
  - [7] E. Yudkowsky and N. Soares, “Functional Decision Theory,” [arXiv:1710.05060](https://arxiv.org/abs/1710.05060) (2017).
  - [8] B. A. Levinstein and N. Soares, “Cheating Death in Damascus,” *The Journal of Philosophy* **117**(5), 237–266 (2020).
  - [9] D. H. Wolpert and G. Benford, “The lesson of Newcomb’s paradox,” *Synthese* **190**, 1637–1646 (2013).
  - [10] D. K. Lewis, “Prisoners’ Dilemma Is a Newcomb Problem,” *Philosophy & Public Affairs* **8**(3), 235–240 (1979).
  - [11] J. H. Sobel, “Some versions of Newcomb’s problem are prisoners’ dilemmas,” *Synthese* **86**(2), 197–208 (1991).
  - [12] C. Schmidt-Petri, “Newcomb’s problem and repeated prisoners’ dilemmas,” *Philosophy of Science* **72**(5), 1160–1173 (2005).
  - [13] G. A. Benford, D. L. Book, and W. A. Newcomb, “The tachyonic antitelephone,” *Phys. Rev. D* **2**, 263–265 (1970).
  - [14] D. Deutsch, “Quantum mechanics near closed timelike lines,” *Phys. Rev. D* **44**, 3197 (1991).
  - [15] S. Lloyd *et al.*, “Closed timelike curves via postselection: theory and experimental test of consistency,” *Phys. Rev. Lett.* **106**, 040403 (2011); [arXiv:1005.2219](https://arxiv.org/abs/1005.2219).
  - [16] C. H. Bennett *et al.*, “Can closed timelike curves or nonlinear quantum mechanics improve quantum state discrimination or help solve hard problems?” *Phys. Rev. Lett.* **103**, 170502 (2009).
  - [17] S. Aaronson, “Quantum computing, postselection, and probabilistic polynomial-time,” *Proc. R. Soc. A* **461**, 3473–3482 (2005); [arXiv:quant-ph/0412187](https://arxiv.org/abs/quant-ph/0412187).
  - [18] S. Aaronson and J. Watrous, “Closed timelike curves make quantum and classical computing equivalent,” *Proc. R. Soc. A* **465**, 631–647 (2009); [arXiv:0808.2669](https://arxiv.org/abs/0808.2669).
  - [19] K. Ji, S. Lloyd, and M. M. Wilde, “Retrocausal capacity of a quantum channel,” *Phys. Rev. Lett.* **136**, 230801 (2026); [arXiv:2509.08965](https://arxiv.org/abs/2509.08965).
  - [20] C. Rovelli, “Relational quantum mechanics,” *Int. J. Theor. Phys.* **35**, 1637–1678 (1996); [arXiv:quant-ph/9609002](https://arxiv.org/abs/quant-ph/9609002).
  - [21] A. Di Biagio and C. Rovelli, “Stable facts, relative facts,” *Found. Phys.* **51**, 30 (2021); [arXiv:2006.15543](https://arxiv.org/abs/2006.15543).
  - [22] Y. Aharonov, P. G. Bergmann, and J. L. Lebowitz, “Time symmetry in the quantum process of measurement,” *Phys. Rev.* **134**, B1410–B1416 (1964).
  - [23] H. Price, “Does time-symmetry imply retrocausality? How the quantum world says ‘maybe’,” *Stud. Hist. Phil. Mod. Phys.* **43**, 75–83 (2012).
  - [24] M. S. Leifer and M. F. Pusey, “Is a time symmetric interpretation of quantum theory possible without retrocausality?” *Proc. R. Soc. A* **473**, 20160607 (2017); [arXiv:1607.07871](https://arxiv.org/abs/1607.07871).
  - [25] A. Kezzyn (dir.), *Stag Hunt*, short film with M. Reilly and S. Lloyd (2023).
  - [26] D. Genkina, G. Chiribella, and L. Hardy, “Optimal probabilistic simulation of quantum channels from the future to the past,” *Phys. Rev. A* **85**, 022330 (2012).
  - [27] H. Reichenbach, *The Direction of Time* (University of California Press, Berkeley, 1956).
  - [28] J.-M. A. Allen, J. Barrett, D. C. Horsman, C. M. Lee, and R. W. Spekkens, “Quantum common causes and quantum causal models,” *Phys. Rev. X* **7**, 031021 (2017); [arXiv:1609.09487](https://arxiv.org/abs/1609.09487).
  - [29] J. F. Nash, “Equilibrium points in  $n$ -person games,” *Proc. Natl. Acad. Sci. USA* **36**, 48–49 (1950).
  - [30] B. Skyrms, *The Stag Hunt and the Evolution of Social Structure* (Cambridge University Press, 2004).
  - [31] J. C. Harsanyi and R. Selten, *A General Theory of Equilibrium Selection in Games* (MIT Press, 1988).
  - [32] J. L. Bermúdez, “Prisoner’s dilemma and Newcomb’s problem: why Lewis’s argument fails,” *Analysis* **73**(3), 423–429 (2013).



- [33] B.-G. Englert, “Fringe visibility and which-way information: an inequality,” *Phys. Rev. Lett.* **77**, 2154–2157 (1996).
- [34] G. Jaeger, A. Shimony, and L. Vaidman, “Two interferometric complementarities,” *Phys. Rev. A* **51**, 54–67 (1995).
- [35] M. Araújo and A. Feix, private communication to Ä. Baumeler and S. Wolf (2014).
- [36] Ä. Baumeler and S. Wolf, “The space of logically consistent classical processes without causal order,” *New J. Phys.* **18**, 013036 (2016); [arXiv:1507.01714](#).
- [37] V. Vilasini and R. Colbeck, “General framework for cyclic and fine-tuned causal models and their compatibility with space-time,” *Phys. Rev. A* **106**, 032204 (2022); [arXiv:2109.12128](#).
- [38] G. T. Horowitz and J. Maldacena, “The black hole final state,” *JHEP* **02** (2004) 008; [arXiv:hep-th/0310281](#).
- [39] S. Lloyd and J. Preskill, “Unitarity of black hole evaporation in final-state projection models,” *JHEP* **08** (2014) 126; [arXiv:1308.4209](#).
- [40] D. Gottesman and J. Preskill, “Comment on ‘The black hole final state’,” *JHEP* **03** (2004) 026; [arXiv:hep-th/0311269](#).
- [41] K. Gödel, “An example of a new type of cosmological solutions of Einstein’s field equations of gravitation,” *Rev. Mod. Phys.* **21**, 447–450 (1949).
- [42] M. S. Morris, K. S. Thorne, and U. Yurtsever, “Wormholes, time machines, and the weak energy condition,” *Phys. Rev. Lett.* **61**, 1446–1449 (1988).
- [43] J. R. Gott, “Closed timelike curves produced by pairs of moving cosmic strings: Exact solutions,” *Phys. Rev. Lett.* **66**, 1126–1129 (1991).
- [44] S. W. Hawking, “Chronology protection conjecture,” *Phys. Rev. D* **46**, 603–611 (1992).
- [45] J. Friedman, M. S. Morris, I. D. Novikov, F. Echeverria, G. Klinkhammer, K. S. Thorne, and U. Yurtsever, “Cauchy problem in spacetimes with closed timelike curves,” *Phys. Rev. D* **42**, 1915–1930 (1990).
- [46] T. A. Brun, J. Harrington, and M. M. Wilde, “Localized closed timelike curves can perfectly distinguish quantum states,” *Phys. Rev. Lett.* **102**, 210402 (2009); [arXiv:0811.1209](#).
- [47] T. A. Brun and M. M. Wilde, “Perfect state distinguishability and computational speedups with postselected closed timelike curves,” *Found. Phys.* **42**, 341–361 (2012); [arXiv:1008.0433](#).
- [48] M. Araújo, P. A. Guérin, and Ä. Baumeler, “Quantum computation with indefinite causal structures,” *Phys. Rev. A* **96**, 052315 (2017); [arXiv:1706.09854](#).
- [49] O. Oreshkov, F. Costa, and Č. Brukner, “Quantum correlations with no causal order,” *Nat. Commun.* **3**, 1092 (2012); [arXiv:1105.4464](#).
- [50] O. Oreshkov and C. Giarmatzi, “Causal and causally separable processes,” *New J. Phys.* **18**, 093020 (2016); [arXiv:1506.05449](#).
- [51] W. L. Craig, “Divine Foreknowledge and Newcomb’s Paradox,” *Philosophia* **17**, 331–350 (1987).
- [52] E. Adlam and C. Rovelli, “Information is physical: cross-perspective links in relational quantum mechanics,” *Philosophy of Physics* **1**(1), 4 (2023); [arXiv:2203.13342](#).
- [53] A. Gibbard and W. L. Harper, “Counterfactuals and two kinds of expected utility,” in *Foundations and Applications of Decision Theory*, Vol. 1, edited by C. A. Hooker, J. J. Leach, and E. F. McClennen (D. Reidel, 1978), pp. 125–162.
- [54] Č. Brukner, “A no-go theorem for observer-independent facts,” *Entropy* **20**, 350 (2018); [arXiv:1804.00749](#).
- [55] D. Frauchiger and R. Renner, “Quantum theory cannot consistently describe the use of itself,” *Nat. Commun.* **9**, 3711 (2018).
- [56] K.-W. Bong, A. Utreras-Alarcón, F. Ghafari, Y.-C. Liang, N. Tischler, E. G. Cavalcanti, G. J. Pryde, and H. M. Wiseman, “A strong no-go theorem on the Wigner’s friend paradox,” *Nat. Phys.* **16**, 1199–1205 (2020).
- [57] M. Smerlak and C. Rovelli, “Relational EPR,” *Found. Phys.* **37**, 427–445 (2007); [arXiv:quant-ph/0604064](#).
- [58] E. G. Cavalcanti, A. Di Biagio, and C. Rovelli, “On the consistency of relative facts,” *Eur. J. Phil. Sci.* **13**, 55 (2023); [arXiv:2305.07343](#).
- [59] J. Lawrence, M. Markiewicz, and M. Żukowski, “Relative facts of relational quantum mechanics are incompatible with quantum mechanics,” *Quantum* **7**, 1015 (2023).
- [60] M. Markiewicz and M. Żukowski, “Relational quantum mechanics with cross-perspective links postulate: an internally inconsistent scheme,” [arXiv:2312.07056](#) (2023).
- [61] C. Rovelli, “Can we travel to the past? Irreversible physics along closed timelike curves,” [arXiv:1912.04702](#) (2019).
- [62] E. H. Simpson, “The interpretation of interaction in contingency tables,” *Journal of the Royal Statistical Society B* **13**(2), 238–241 (1951).
- [63] D. A. Meyer, “Quantum strategies,” *Phys. Rev. Lett.* **82**, 1052–1055 (1999).
- [64] J. Eisert, M. Wilkens, and M. Lewenstein, “Quantum games and quantum strategies,” *Phys. Rev. Lett.* **83**, 3077–3080 (1999).
- [65] E. W. Piotrowski and J. Śladowski, “An invitation to quantum game theory,” *Int. J. Theor. Phys.* **42**, 1089–1099 (2003).
- [66] E. W. Piotrowski and J. Śladowski, “Quantum solution to the Newcomb’s paradox,” [arXiv:quant-ph/0202074](#) (2002).
- [67] D. P. Chi and K. Jeong, “Quantum Solution to the Extended Newcomb’s Paradox,” [arXiv:quant-ph/0511097](#) (2005).
- [68] E. G. Cavalcanti, “Causation, decision theory, and Bell’s theorem: a quantum analogue of the Newcomb problem,” *Brit. J. Phil. Sci.* **61**, 569–597 (2010); [arXiv:0911.2500](#).
- [69] Y. Aharonov, D. Z. Albert, and L. Vaidman, “How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100,” *Phys. Rev. Lett.* **60**, 1351–1354 (1988).
- [70] F. Echeverria, G. Klinkhammer, and K. S. Thorne, “Billiard balls in wormhole spacetimes with closed timelike curves: Classical theory,” *Phys. Rev. D* **44**, 1077 (1991).
- [71] C. Ferradini, V. Gitton, and V. Vilasini, “Cyclic functional causal models beyond unique solvability with a graph separation theorem,” [arXiv:2502.04171](#) (2025).
- [72] Z. Liu and O. Oreshkov, “Parity erasure: a foundational principle for indefinite causal order,” [arXiv:2512.08635](#) (2025).